Software Engineering

Digital Assignment – 3 / Review – 3

Project Title: Scrollhalla: E-Reader RSS Feed

Testing · Version Management · System Building · Final Project Report

By Tejas Jaiswal (24BAI1264) &

Dhananjay Shankar (24BAI1298)

Submitted to Dr. Alok Chauhan

GitHub: **https://github.com/tejasjaiswal2024-art/Scrollhalla**

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1. Integration, Regression and Mutation Testing

Until this review the repository had no automated tests. For DA3 a four-level test suite was built around the backend's existing layered structure (routes → controllers → services → domain models → infrastructure adapters), plus a frontend unit suite. All suites run with one command per level and in CI on every push.

116	92.3%	86.2% → 96.5%	10
automated tests (111 backend + 5 frontend)	backend line coverage (87.1% branch)	mutation score (Stryker, 489 mutants)	regression snapshots (API contract + golden master)

Level	Tool	What it exercises	Tests	Command
Unit	Jest + ts-jest	Task, BacklogItem, SmartBacklogService, SprintRiskService, RssParser, AtomParser, FeedParserFactory, CacheService fallback, JWT middleware — each in isolation, network stubbed.	74	`npm run test:unit`
Integration	Jest + supertest	The real Express app end-to-end through router, controllers, services and models. Only Layer-5 adapters (Prisma, Redis, outbound HTTP) are replaced by in-memory fakes.	23	`npm run test:integration`
Regression	Jest snapshots	Status code + response shape of every endpoint, the ranking of the seed backlog (golden master), and previously-identified edge cases.	14	`npm run test:regression`
Mutation	Stryker Mutator	Injects 489 small faults into models, services, parsers, middleware and controllers and checks that the suite above detects each one.	489 mutants	`npm run test:mutation`
Frontend unit	Vitest	OPML 2.0 export: envelope, one outline per feed, category default, XML escaping of all five special characters.	5	`npm test` (frontend)

1.1 Unit testing

Unit tests pin the business rules that the rest of the system depends on: the default values of a `Task`, the MoSCoW weight table, the exact Smart Backlog scoring formula, the severity classification and risk thresholds of the Sprint Risk service, and the normalisation each parser performs. The RSS client is replaced by a shared stub so no test touches the network.

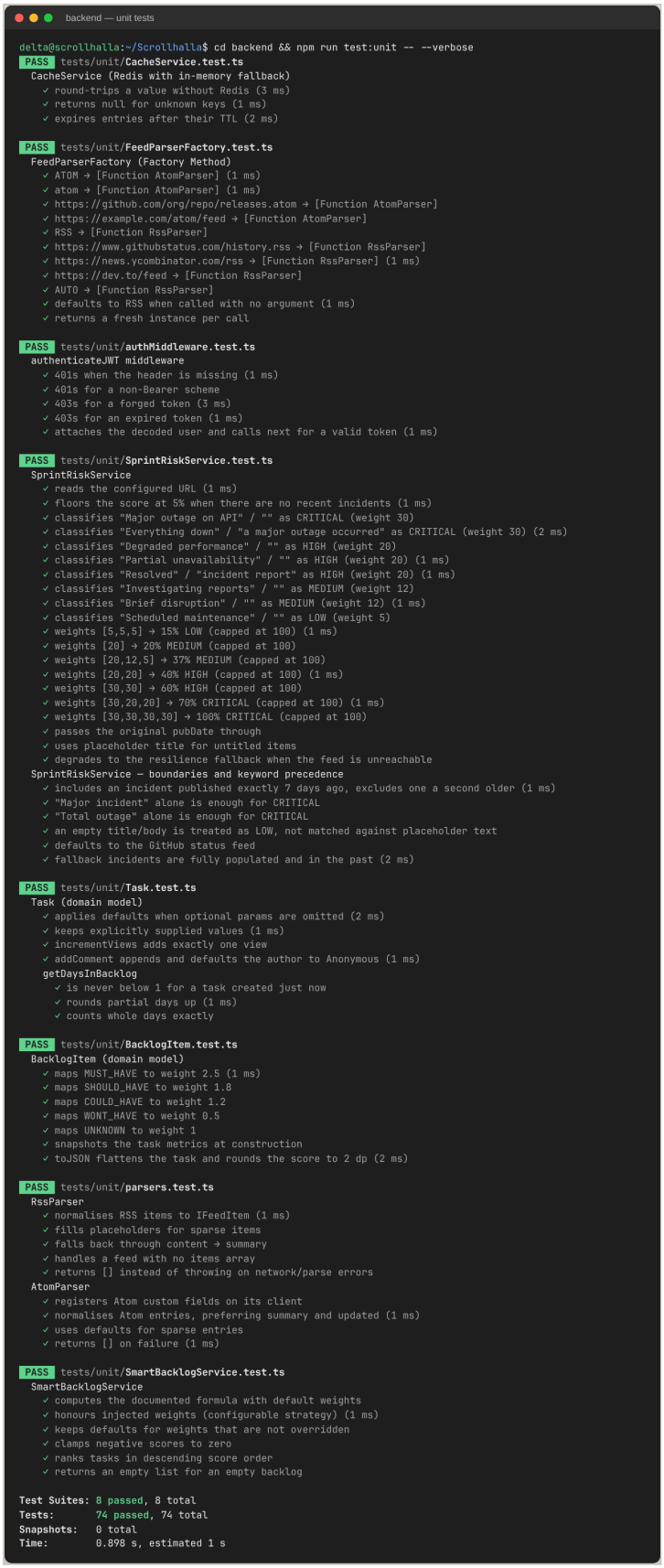


Figure 1 — Unit suite: 74 tests across 8 suites, all passing.

1.2 Integration testing

The integration suite drives the real application over HTTP with `supertest`. The composition root in `routes/api.ts` is left untouched, so the test also verifies that dependency injection wires the controllers correctly — for instance, that a task created through `/api/tasks` is visible to the Smart Backlog and Team controllers, which share the same `TaskController` instance. Five integration scenarios are covered:

- **Task CRUD flow** — create, read (view counter increments), update, comment, delete, and 404 on every verb afterwards.
- **Cross-controller** — a high-engagement task rises to the top of `/api/smart-backlog`; reassignment changes team workload.
- **Feed pipeline** — cold cache triggers on-demand harvest through Factory → Parser → Cache; the second request is a `REDIS_CACHE_HIT`; one failing publisher does not break the timeline.
- **Sprint risk** — the endpoint degrades to the resilience fallback when the status feed is offline.
- **Auth flow** — register → duplicate refused → login → protected profile with bcrypt and JWT; identical 401 for wrong password and unknown user.

backend — integration tests

```
delta@scrollhalla:~/Scrollhalla$ cd backend && npm run test:integration -- --verbose
 PASS  tests/integration/api.int.test.ts
  Health & routing
    ✓ GET /health reports UP (22 ms)
    ✓ unknown routes return the JSON 404 envelope (10 ms)
  Task CRUD flow (TaskController ⇄ Task)
    ✓ lists the seeded tasks (3 ms)
    ✓ filters by MoSCoW alone (7 ms)
    ✓ filters by status and MoSCoW (3 ms)
    ✓ rejects a task without title/description (9 ms)
    ✓ creates a task with defaults (3 ms)
    ✓ reading a task increments its view count (5 ms)
    ✓ updates only supplied fields (2 ms)
    ✓ adds a comment, rejects an empty one (3 ms)
    ✓ deletes the task, then 404s on every verb (7 ms)
  Cross-controller integration (shared TaskController via DI)
    ✓ a newly created high-engagement task rises to the top of the smart backlog (136 ms)
    ✓ team workload reflects reassignment made through the task API (6 ms)
  Feed pipeline (FeedController → CronWorker → Factory → Parsers → Cache)
    ✓ cold cache: harvests on demand, merges feeds newest-first, then serves from cache (4 ms)
    ✓ one failing source does not break the timeline (2 ms)
  Sprint risk endpoint
    ✓ returns the fallback assessment when the status feed is offline (2 ms)
  Auth flow (AuthController ⇄ bcrypt ⇄ JWT ⇄ middleware)
    ✓ validates required fields (4 ms)
    ✓ registers, never echoing the password hash (81 ms)
    ✓ refuses duplicate registration (1 ms)
    ✓ rejects wrong password and unknown user with the same 401 (74 ms)
    ✓ logs in and reaches the protected profile (72 ms)
    ✓ blocks the profile without a token (1 ms)
  Team workload is computed per member
    ✓ each member only counts their own tasks (4 ms)

Test Suites: 1 passed, 1 total
Tests:       23 passed, 23 total
Snapshots:   0 total
Time:        0.918 s, estimated 1 s
```

Figure 2 — Integration suite: 23 tests across the full request path.

Listing 1 — Integration tests replace only the infrastructure adapters (backend/tests/integration/api.int.test.ts)

```
// Integration: real Express app → router → controllers → services → models.
// Only Layer-5 adapters (Prisma, Redis, outbound HTTP) are replaced.
jest.mock('../../src/config/prisma', () ⇒
require('../helpers/mockInfra').prismaModuleMock());
jest.mock('../../src/config/redis',  () ⇒
require('../helpers/mockInfra').cacheModuleMock());
jest.mock('rss-parser', () ⇒ require('../helpers/rssParserMock').rssParserModuleMock());

it('cold cache: harvests on demand, merges feeds newest-first, then serves from cache', async
() ⇒ {
  const first = await request(app).get('/api/feed');
  expect(first.body).toMatchObject({ success: true, source: 'ON_DEMAND_HARVEST', count: 3 });
  const second = await request(app).get('/api/feed');
  expect(second.body.source).toBe('REDIS_CACHE_HIT');
  expect(parseURL).toHaveBeenCalledTimes(3);    // no second harvest
});
```

1.3 Regression testing

The regression pack protects behaviour that users and the frontend already rely on. REG-01 snapshots the status code and the *shape* of every endpoint's response — volatile values such as IDs and timestamps are reduced to their type so the snapshot only changes when the contract changes. REG-02 is a golden master of the seeded backlog ranking. REG-03 pins edge cases: the documented `storyPoints: 0 → 3` coercion, the 100% cap on team utilisation, non-enumerating login errors, and malformed JSON being handled by the global error handler.

backend — regression tests

```
delta@scrollhalla:~/Scrollhalla$ cd backend && npm run test:regression -- --verbose
 PASS  tests/regression/api-contract.reg.test.ts
  REG-01 response contracts
    ✓ GET /health (33 ms)
    ✓ GET /api/tasks (15 ms)
    ✓ GET /api/tasks/TASK-101 (4 ms)
    ✓ GET /api/smart-backlog (6 ms)
    ✓ GET /api/team (6 ms)
    ✓ GET /api/feed (4 ms)
    ✓ GET /api/sprint-risk (5 ms)
    ✓ GET /api/tasks/TASK-0000 (2 ms)
    ✓ GET /api/auth/me (8 ms)
  REG-02 seed backlog ranking (golden master)
    ✓ orders the seeded tasks identically to the released build (5 ms)
  REG-03 previously-fixed / easily-broken behaviours
    ✓ storyPoints of 0 is coerced to the default 3 (documented quirk, not silently changed) (15 ms)
    ✓ team utilisation is capped at 100% (12 ms)
    ✓ auth error messages do not reveal whether an email is registered (4 ms)
    ✓ malformed JSON bodies are handled by the global error handler, not crash the process (4 ms)

Test Suites: 1 passed, 1 total
Tests:       14 passed, 14 total
Snapshots:   10 passed, 10 total
Time:        0.657 s, estimated 1 s
```

Figure 3 — Regression pack: 14 tests, 10 snapshots matched.

Listing 2 — Contract snapshot that ignores volatile values (backend/tests/regression/api-contract.reg.test.ts)

```
/** Replace volatile values (ids, timestamps, latencies) with their type so snapshots are
stable. */
const shape = (v: any): any => {
  if (Array.isArray(v)) return v.length ? [shape(v[0])] : [];
  if (v && typeof v === 'object') return Object.fromEntries(Object.keys(v).sort().map(k =>
[k, shape(v[k])]));
  return typeof v;
};
it.each([['GET', '/health'], ['GET', '/api/tasks'], ['GET', '/api/smart-backlog'], /* … 9
endpoints */])('%s %s', async (_m, url) => {
  const r = await request(app).get(url);
  expect({ status: r.status, body: shape(r.body) }).toMatchSnapshot();
});
```

1.4 Coverage

backend — coverage

```
delta@scrollhalla:~/Scrollhalla$ cd backend && npm run test:coverage
 PASS  tests/integration/api.int.test.ts
 PASS  tests/regression/api-contract.reg.test.ts
 PASS  tests/unit/CacheService.test.ts
 PASS  tests/unit/authMiddleware.test.ts
 PASS  tests/unit/FeedParserFactory.test.ts
 PASS  tests/unit/SprintRiskService.test.ts
 PASS  tests/unit/Task.test.ts
 PASS  tests/unit/SmartBacklogService.test.ts
 PASS  tests/unit/parsers.test.ts
 PASS  tests/unit/BacklogItem.test.ts
----------------------------|---------|----------|---------|---------|-----------------------------------
File                        | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
----------------------------|---------|----------|---------|---------|-----------------------------------
All files                   |    91.2 |    87.07 |    92.2 |    92.3 |
 src                        |     100 |      100 |     100 |     100 |
  app.ts                    |     100 |      100 |     100 |     100 |
 src/config                 |   66.66 |    42.85 |   83.33 |   67.64 |
  redis.ts                  |   66.66 |    42.85 |   83.33 |   67.64 | 24-25,31,37-41,53-57
 src/controllers            |   90.85 |    83.72 |     100 |   92.54 |
  AuthController.ts         |   83.01 |       80 |     100 |   83.01 | 60-61,108-109,120-121,141-142,150
  FeedController.ts         |   88.23 |      100 |     100 |   88.23 | 38-39
  SmartBacklogController.ts |     100 |      100 |     100 |     100 |
  SprintRiskController.ts   |    87.5 |      100 |     100 |    87.5 | 19
  TaskController.ts         |   94.73 |    83.33 |     100 |     100 | 85-86,88,90
  TeamController.ts         |     100 |      100 |     100 |     100 |
 src/middleware             |     100 |       80 |     100 |     100 |
  authMiddleware.ts         |     100 |       80 |     100 |     100 | 4
 src/models                 |     100 |      100 |     100 |     100 |
  BacklogItem.ts            |     100 |      100 |     100 |     100 |
  Task.ts                   |     100 |      100 |     100 |     100 |
 src/parsers                |     100 |    93.33 |     100 |     100 |
  AtomParser.ts             |     100 |    86.36 |     100 |     100 | 21-22
  BaseParser.ts             |     100 |      100 |     100 |     100 |
  FeedParserFactory.ts      |     100 |      100 |     100 |     100 |
  RssParser.ts              |     100 |      100 |     100 |     100 |
 src/routes                 |     100 |      100 |     100 |     100 |
  api.ts                    |     100 |      100 |     100 |     100 |
 src/services               |     100 |    94.87 |     100 |     100 |
  SmartBacklogService.ts    |     100 |      100 |     100 |     100 |
  SprintRiskService.ts      |     100 |    93.54 |     100 |     100 | 34,82
 src/workers                |   68.42 |       60 |    37.5 |   71.42 |
  feedCronWorker.ts         |   68.42 |       60 |    37.5 |   71.42 | 29,34-35,50,68-78
----------------------------|---------|----------|---------|---------|-----------------------------------

============================== Coverage summary ==============================
Statements   : 91.2% ( 415/455 )
Branches     : 87.07% ( 155/178 )
Functions    : 92.2% ( 71/77 )
Lines        : 92.3% ( 396/429 )
==============================================================================

Test Suites: 10 passed, 10 total
Tests:       111 passed, 111 total
Snapshots:   10 passed, 10 total
Time:        2.103 s
```

Figure 4 — Istanbul coverage across all backend source files.

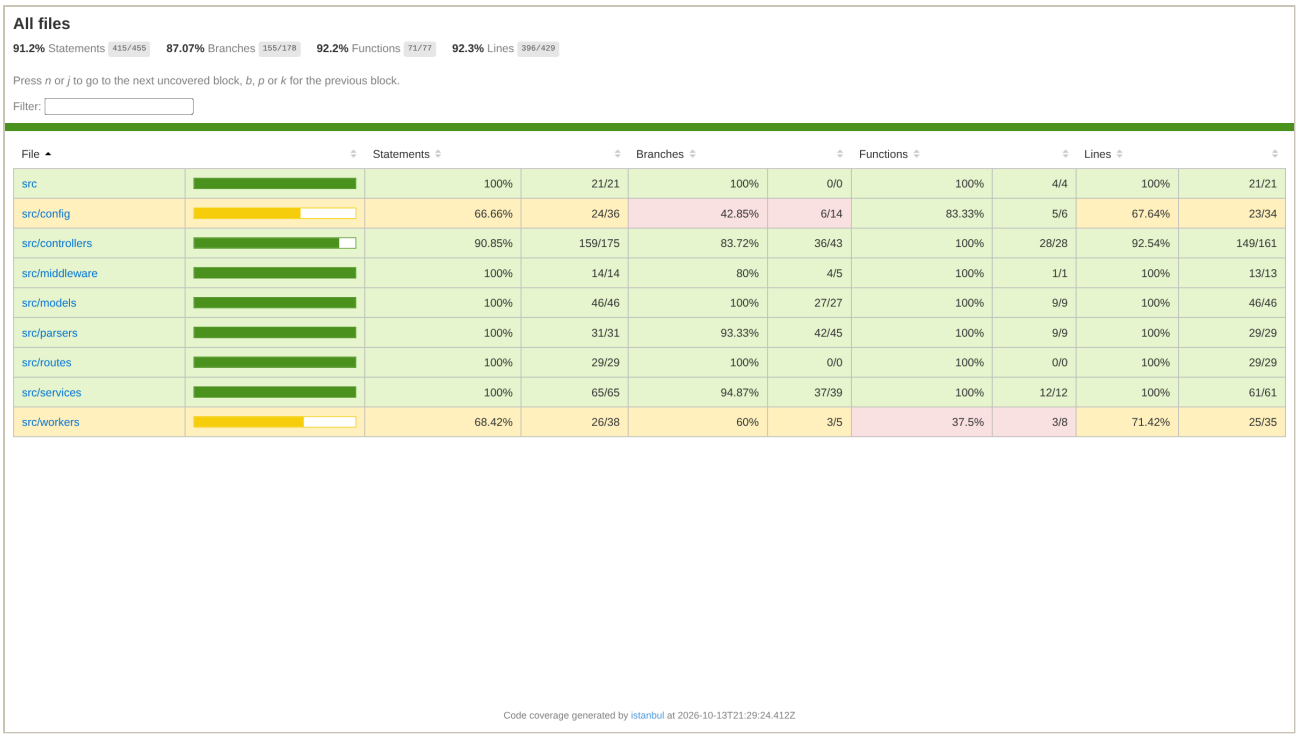


Figure 5 — HTML coverage report (uploaded as a CI artifact on every run).

1.5 Mutation testing

Line coverage only shows that code *ran*; it does not show that a test would fail if the code were wrong. Stryker answers that question by generating mutants — `⩾` turned into `>`, `||` into `&&`, a string emptied, a branch forced to `true` — and re-running the suite against each one. A mutant that no test notices *survives* and marks a weak spot.

Stryker — baseline run

```
delta@scrollhalla:~/Scrollhalla$ cd backend && npm run test:mutation
$ npm run test:mutation   # baseline
-------------------------|------------------|----------|----------|-----------|----------|----------|
                         | % Mutation score |          |          |           |          |          |
File                     |  total | covered | # killed | # timeout | # survived | # no cov | # errors |
-------------------------|--------|---------|----------|----------|-----------|----------|----------|
All files                |  86.17 |   86.87 |      423 |         7 |         65 |        4 |        4 |
 controllers             |  89.26 |   89.26 |      127 |         6 |         16 |        0 |        0 |
  TaskController.ts      |  90.98 |   90.98 |      116 |         5 |         12 |        0 |        0 |
  TeamController.ts      |  75.00 |   75.00 |       11 |         1 |          4 |        0 |        0 |
 middleware              |  73.91 |   77.27 |       16 |         1 |          5 |        1 |        0 |
  authMiddleware.ts      |  73.91 |   77.27 |       16 |         1 |          5 |        1 |        0 |
 models                  | 100.00 |  100.00 |       56 |         0 |          0 |        0 |        0 |
  BacklogItem.ts         | 100.00 |  100.00 |        9 |         0 |          0 |        0 |        0 |
  Task.ts                | 100.00 |  100.00 |       47 |         0 |          0 |        0 |        0 |
 parsers                 |  82.11 |   83.47 |      101 |         0 |         20 |        2 |        0 |
  AtomParser.ts          |  92.59 |   96.15 |       50 |         0 |          2 |        2 |        0 |
  BaseParser.ts          |   0.00 |    0.00 |        0 |         0 |          1 |        0 |        0 |
  FeedParserFactory.ts   |  46.43 |   46.43 |       13 |         0 |         15 |        0 |        0 |
  RssParser.ts           |  95.00 |   95.00 |       38 |         0 |          2 |        0 |        0 |
 services                |  83.11 |   83.67 |      123 |         0 |         24 |        1 |        4 |
  SmartBacklogService.ts | 100.00 |  100.00 |       24 |         0 |          0 |        0 |        4 |
  SprintRiskService.ts   |  79.84 |   80.49 |       99 |         0 |         24 |        1 |        0 |
-------------------------|--------|---------|----------|----------|-----------|----------|----------|
MutationTestReportHelper Final mutation score of 86.17 is greater than or equal to break threshold 60
```

Figure 6 — Baseline run: 86.17% mutation score, 65 surviving mutants.

The baseline run surfaced four real problems, each fixed in a separate commit:

Surviving mutants	What they revealed	Action
15 in `FeedParserFactory` RSS condition	Every mutation of the RSS branch survived because the branch was **unreachable**: whatever it matched already fell through to the RSS default.	Refactored the factory to a single Atom check (commit `5e71a78`).
4 in `TaskController` moscow filter	The only filter test combined status and MoSCoW; every `IN_PROGRESS` seed task is `MUST_HAVE`, so the MoSCoW filter was never actually tested.	Added a MoSCoW-only filter test.
5 in `authMiddleware`	Tests checked status codes but never the response body, so emptying the error message went unnoticed.	Assert full 401 and 403 bodies.
~20 in `SprintRiskService`	The 7-day boundary (`⩽` vs `<`), keyword precedence ("major" alone), and fallback incident fields were unasserted.	Boundary test with fake timers; keyword and fallback assertions.

Stryker — after test hardening

```
delta@scrollhalla:~/Scrollhalla$ cd backend && npm run test:mutation
$ npm run test:mutation   # final
-------------------------|-------------------|----------|----------|-----------|----------|----------|
                         | % Mutation score |          |          |           |          |          |
File                     |  total | covered | # killed | # timeout | # survived | # no cov | # errors |
-------------------------|--------|---------|----------|----------|-----------|----------|----------|
All files                |  96.49 |   97.30 |      461 |         7 |         13 |        4 |        4 |
 controllers             |  97.99 |   97.99 |      140 |         6 |          3 |        0 |        0 |
  TaskController.ts      |  97.74 |   97.74 |      125 |         5 |          3 |        0 |        0 |
  TeamController.ts      | 100.00 |  100.00 |       15 |         1 |          0 |        0 |        0 |
 middleware              |  95.65 |  100.00 |       21 |         1 |          0 |        1 |        0 |
  authMiddleware.ts      |  95.65 |  100.00 |       21 |         1 |          0 |        1 |        0 |
 models                  | 100.00 |  100.00 |       56 |         0 |          0 |        0 |        0 |
  BacklogItem.ts         | 100.00 |  100.00 |        9 |         0 |          0 |        0 |        0 |
  Task.ts                | 100.00 |  100.00 |       47 |         0 |          0 |        0 |        0 |
 parsers                 |  92.66 |   94.39 |      101 |         0 |          6 |        2 |        0 |
  AtomParser.ts          |  92.59 |   96.15 |       50 |         0 |          2 |        2 |        0 |
  BaseParser.ts          |   0.00 |    0.00 |        0 |         0 |          1 |        0 |        0 |
  FeedParserFactory.ts   |  92.86 |   92.86 |       13 |         0 |          1 |        0 |        0 |
  RssParser.ts           |  95.00 |   95.00 |       38 |         0 |          2 |        0 |        0 |
 services                |  96.62 |   97.28 |      143 |         0 |          4 |        1 |        4 |
  SmartBacklogService.ts | 100.00 |  100.00 |       24 |         0 |          0 |        0 |        4 |
  SprintRiskService.ts   |  95.97 |   96.75 |      119 |         0 |          4 |        1 |        0 |
-------------------------|--------|---------|----------|----------|-----------|----------|----------|
MutationTestReportHelper Final mutation score of 96.49 is greater than or equal to break threshold 60
```

Figure 7 — After hardening: 96.49% mutation score (461 killed, 7 timeout, 13 survived).

Remaining survivors are equivalent mutants, meaning changes that cannot alter observable behaviour, so no test can kill them. Examples: `status || TaskStatus.TO_DO` in the controller (the `Task` constructor applies the same default); `item.pubDate || new Date()` (items without a date are already filtered out above); the unused `parserName` field; and log-message strings. They are documented here rather than suppressed.

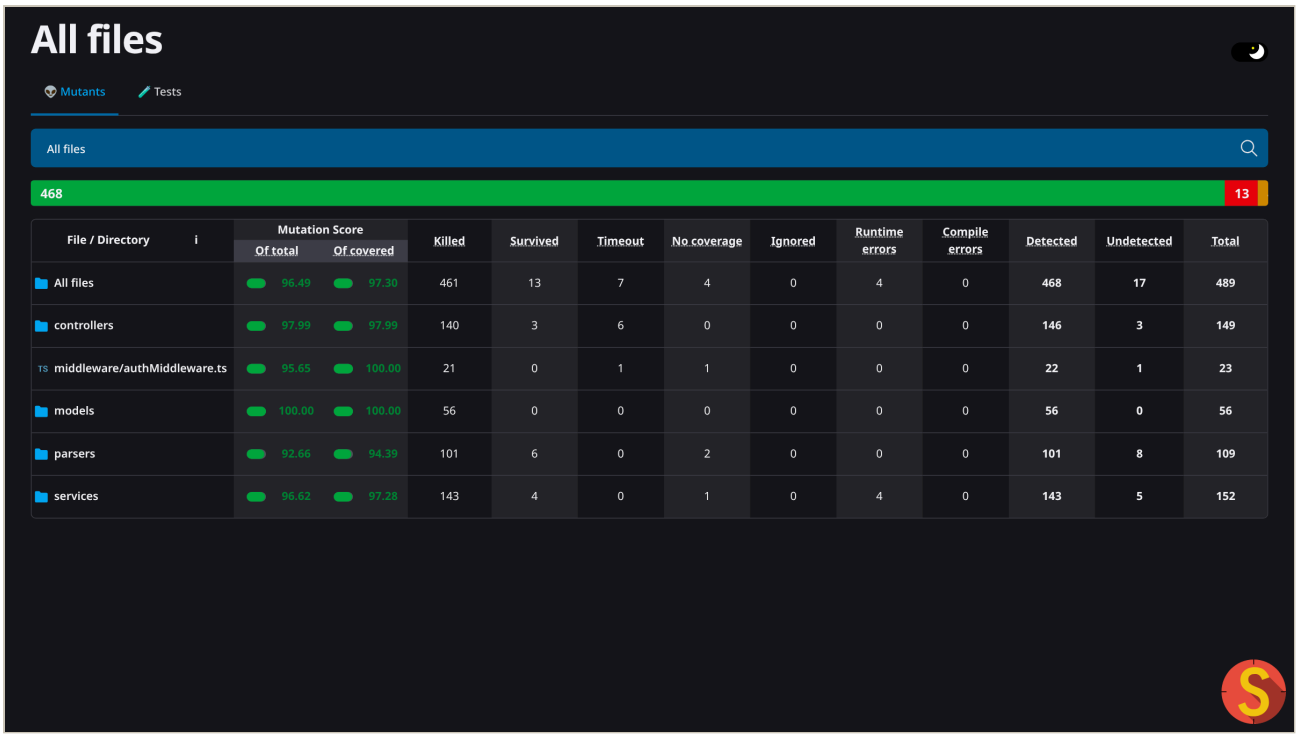


Figure 8 — Stryker HTML report (CI artifact).

1.6 Frontend tests

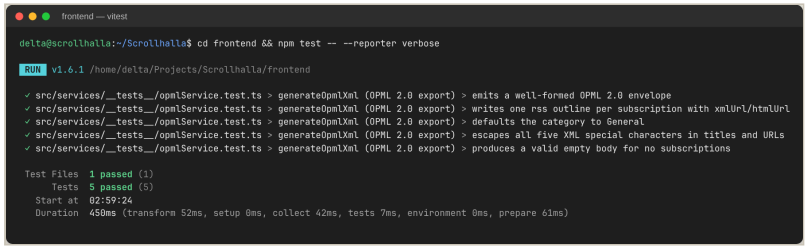


Figure 9 — Vitest suite for the OPML 2.0 export service.

2. Version Management and System Building

2.1 Branching strategy

The project follows **GitHub Flow**, as set up in DA1: `main` holds the stable, deployable version and every unit of work lives on a `feature/*` branch merged through a pull request. Each review has its own branch (`feature/da2-design`, `feature/da3-testing`) next to the functional branches (`feature/onboarding-rss-enhancements`, `feature/opml-feed-exporter`). Commits use the **Conventional Commits** prefixes `feat`, `fix`, `test`, `refactor`, `docs` and `ci`, so the history reads as a changelog. Releases are marked with semantic-version tags.

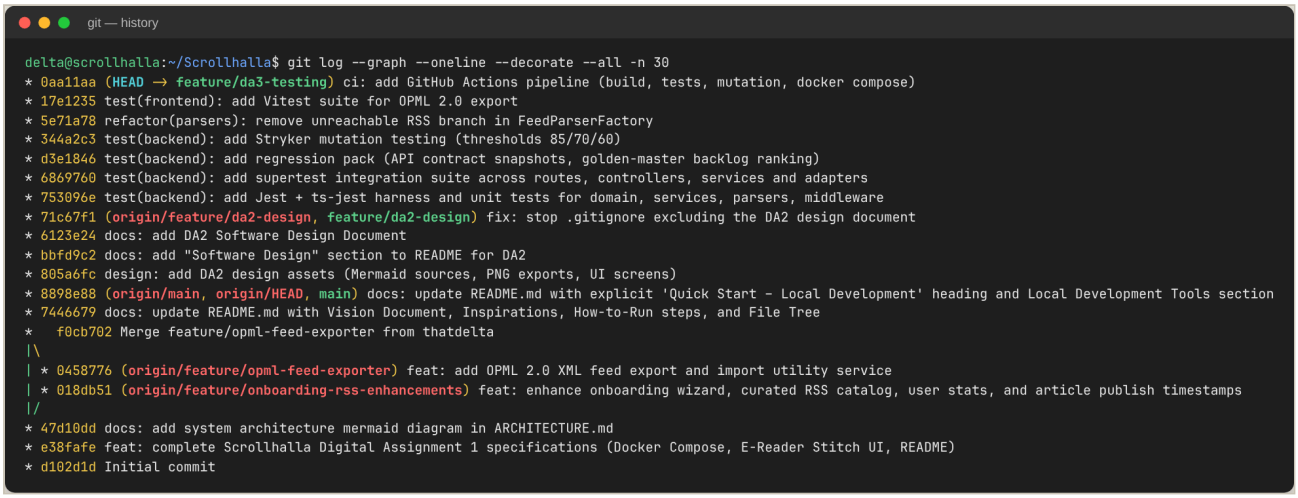


Figure 10 — Commit graph: feature branches merged into main; DA3 work as small, single-purpose commits.

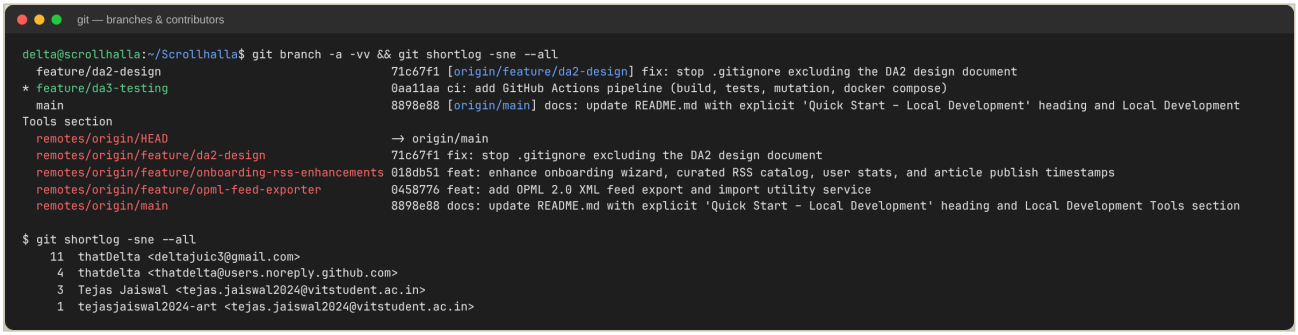


Figure 11 — Local and remote branches with upstream tracking, and contributors.



Figure 12 — Annotated release tags, one per review milestone.

2.2 System building

The system is built at three levels, and each one is reproducible from a clean checkout:

1. **Component builds.** The backend runs `npm run build`, which is `prisma generate` followed by `tsc`, the type-check gate. The frontend runs `npm run build`, which is a Vite production bundle of 215 kB (65 kB gzip).
2. **Container images.** The backend uses a multi-stage Dockerfile, with a build stage and a slim runtime stage on `node:20-alpine`. The frontend Vite bundle is served by Nginx, which also reverse-proxies `/api`.
3. **System composition.** `docker compose up --build` starts four services: `postgres:15`, `redis:7`, the backend and the frontend. They share one network, and health-checked dependencies control the start order.

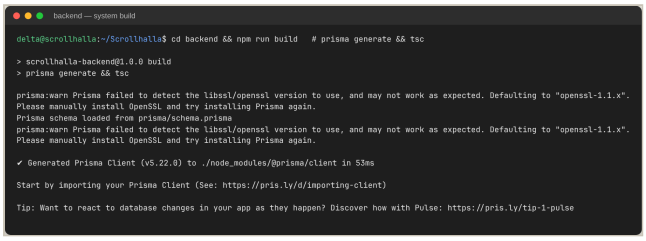


Figure 13 — Backend build (Prisma client + tsc).

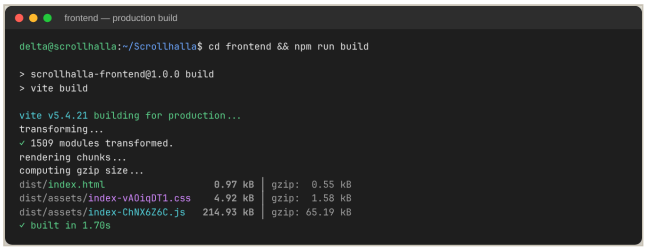


Figure 14 — Frontend production build.

2.3 Continuous integration

A GitHub Actions workflow (`.github/workflows/ci.yml`) runs on every push to `main` and `feature/**`, on every pull request and on every `v*` tag. It has four jobs:

- **backend** builds the code, then runs the unit, integration and regression tests and coverage.
- **mutation** runs Stryker after the backend job and fails the pipeline if the score falls below 60%.
- **frontend** runs the Vitest tests and the production build in parallel with the backend job.
- **docker** builds the whole compose stack once both the backend and frontend jobs pass.

The coverage report, the mutation report and the frontend `dist/` are uploaded as artifacts.

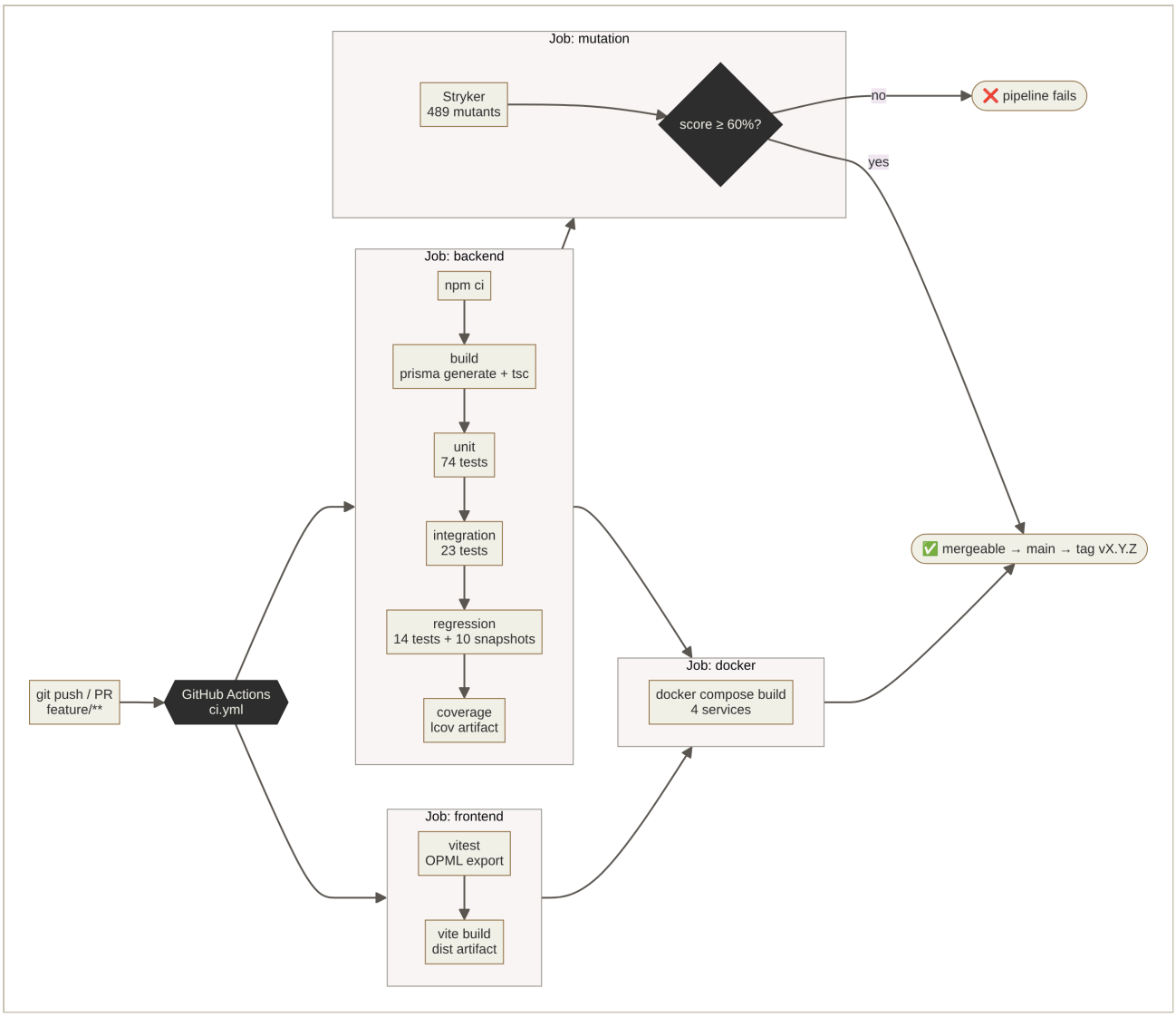


Figure 15 — CI pipeline: build → test levels → mutation gate → system build.

Listing 3 — Backend job of the CI pipeline (.github/workflows/ci.yml, excerpt)

```
  backend:
    runs-on: ubuntu-latest
    defaults: { run: { working-directory: backend } }
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-node@v4
        with: { node-version: 20, cache: npm, cache-dependency-path: backend/package-
lock.json }
      - run: npm ci
      - run: npm run build           # prisma generate + tsc (type-check gate)
      - run: npm run test:unit
      - run: npm run test:integration
      - run: npm run test:regression -- --ci
      - run: npm run test:coverage -- --ci
```

3. Screenshots of Developed Functionalities

All screenshots below come from the running system. The Vite frontend talks to the Express backend, which uses real PostgreSQL and Redis instances, and the articles shown are live items harvested from the DEV Community and Hacker News RSS feeds at the time of capture.



Figure 16 — Onboarding: reader profile, interest tags, seed channels (US #3).

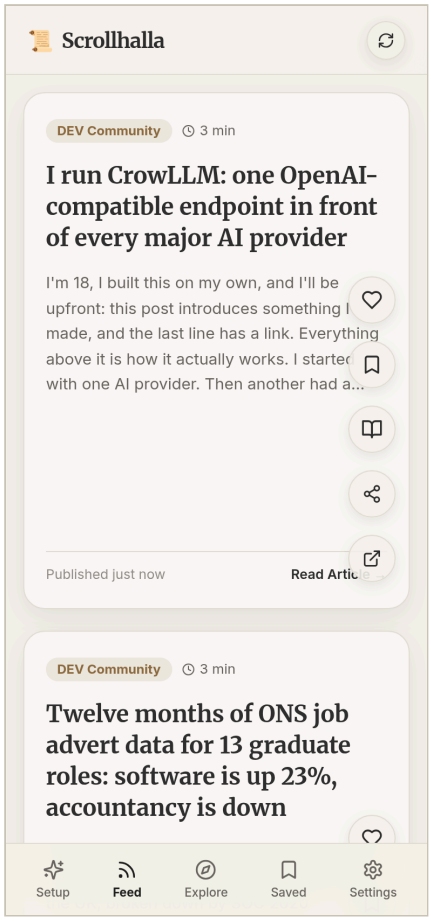


Figure 17 — Reels-style vertical feed, mobile viewport, bottom navigation (US #4, #18, #21).

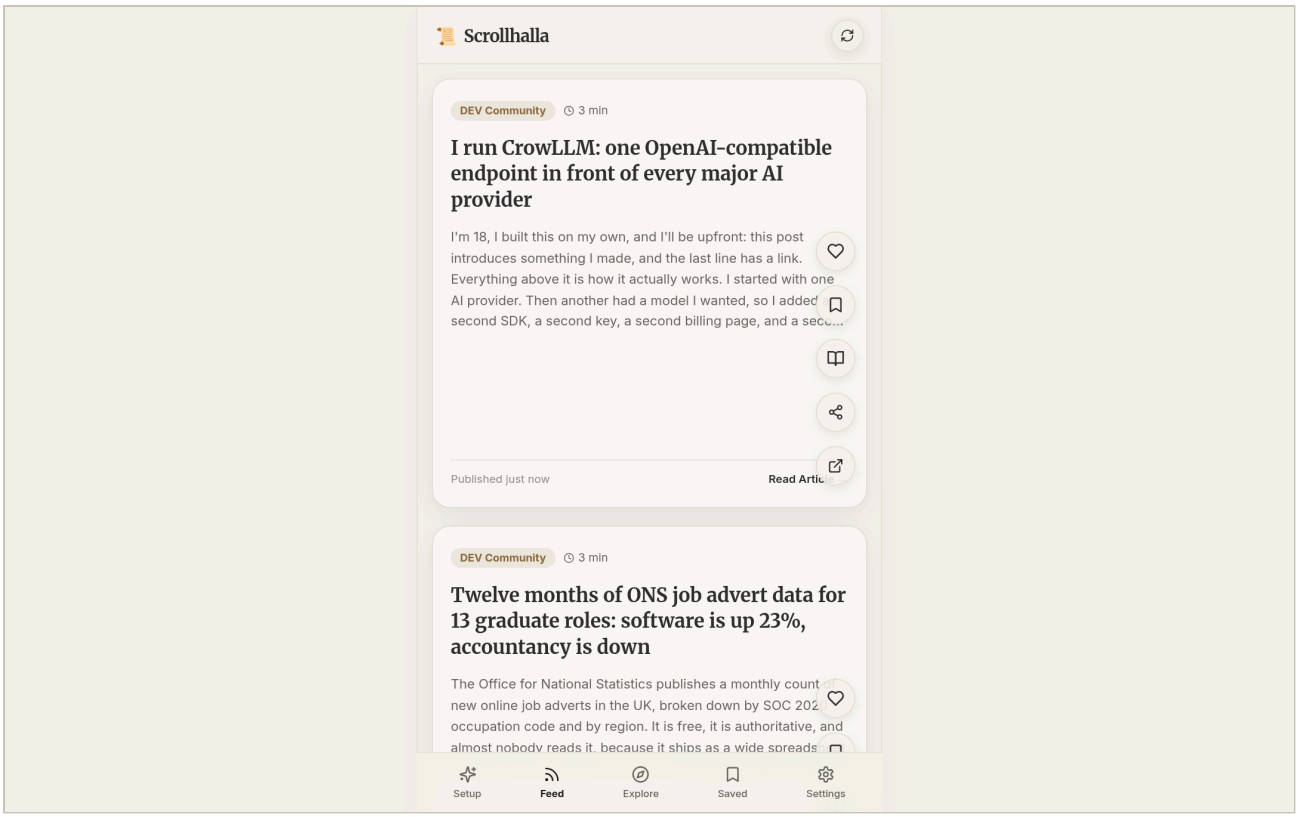


Figure 18 — Timeline served from the Redis cache, with overlay actions: Like, Save, Focus Reader, Share, Open (US #4, #7, #8, #25).

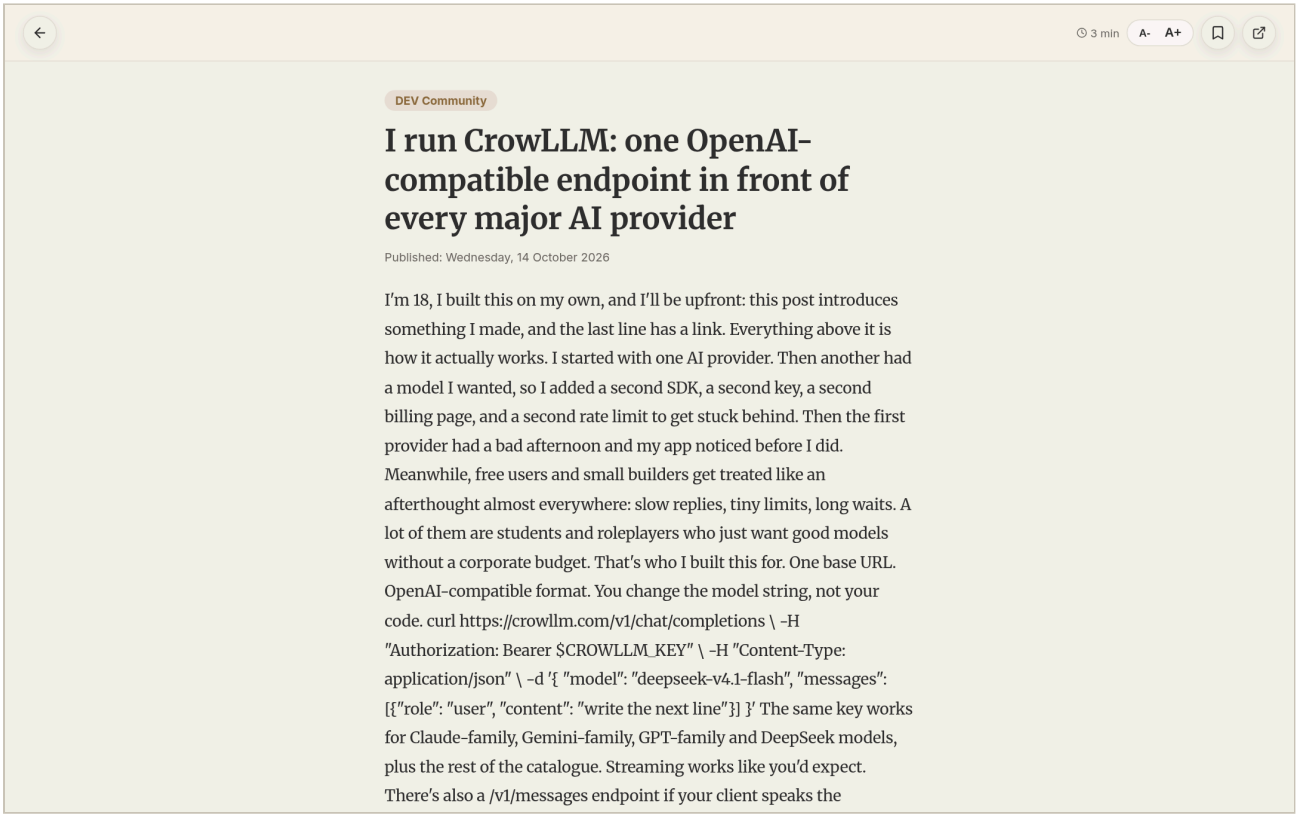


Figure 19 — Distraction-free reader view: serif body text, A-/A+ font size, bookmark (US #6, #12, #17).

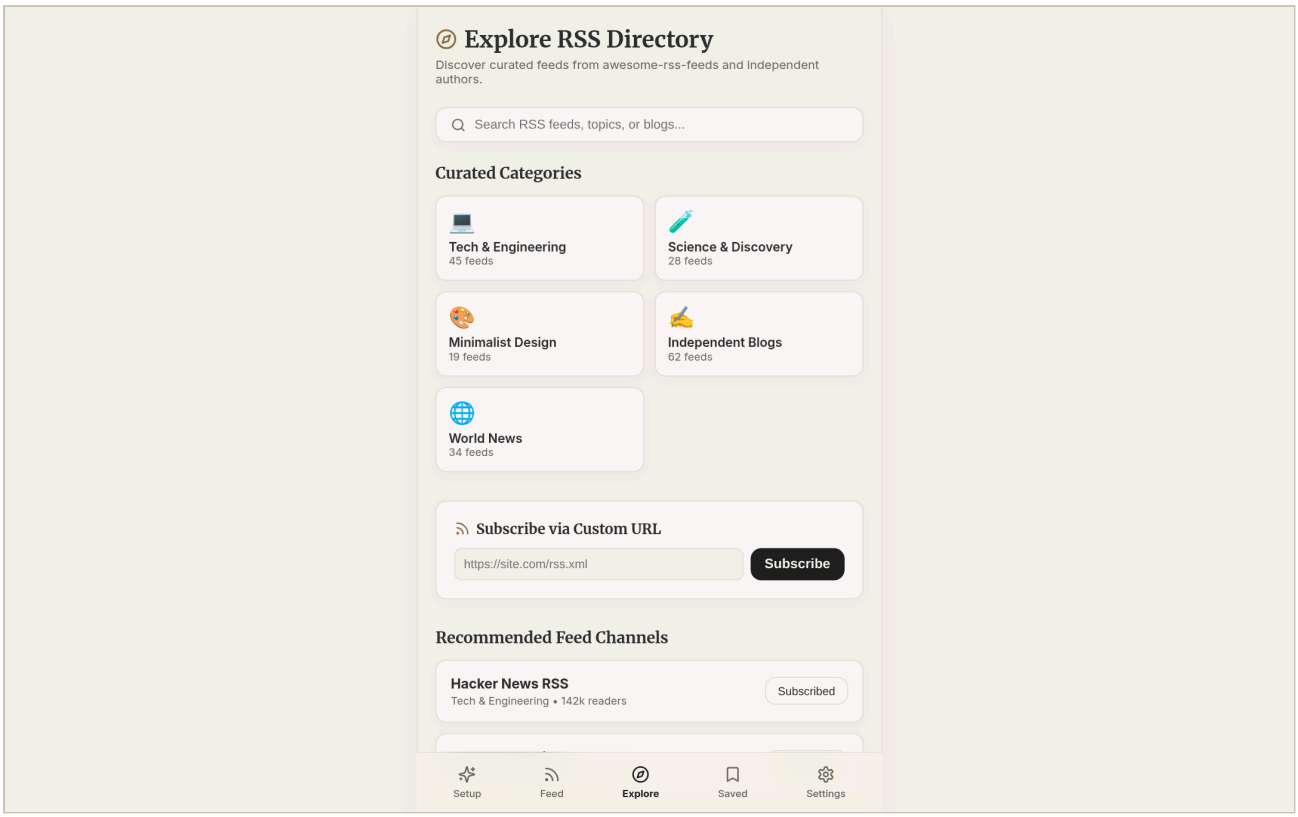


Figure 20 — Explore directory: curated categories, subscribe by custom RSS URL, recommended channels (US #10, #11).

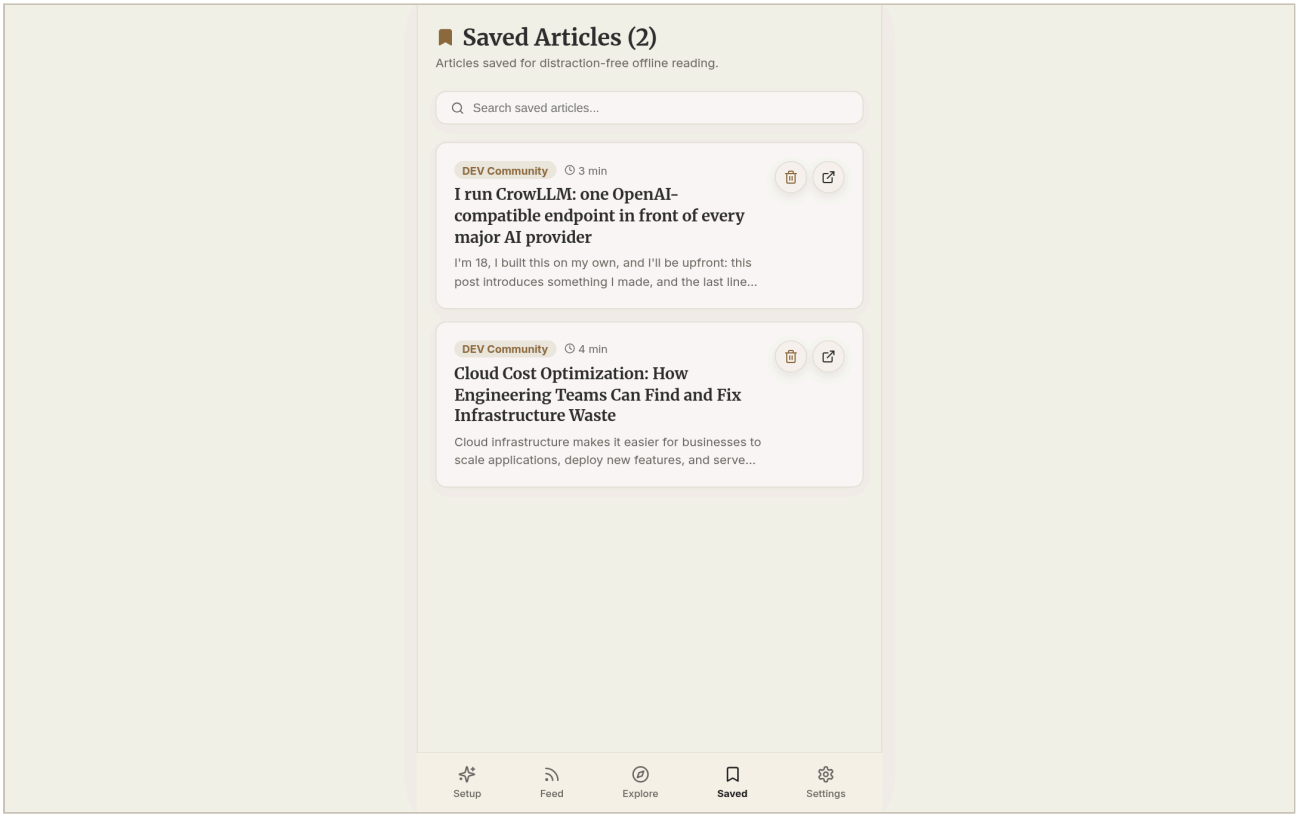


Figure 21 — Saved articles: bookmarked from the timeline, searchable, removable (US #9, #24).

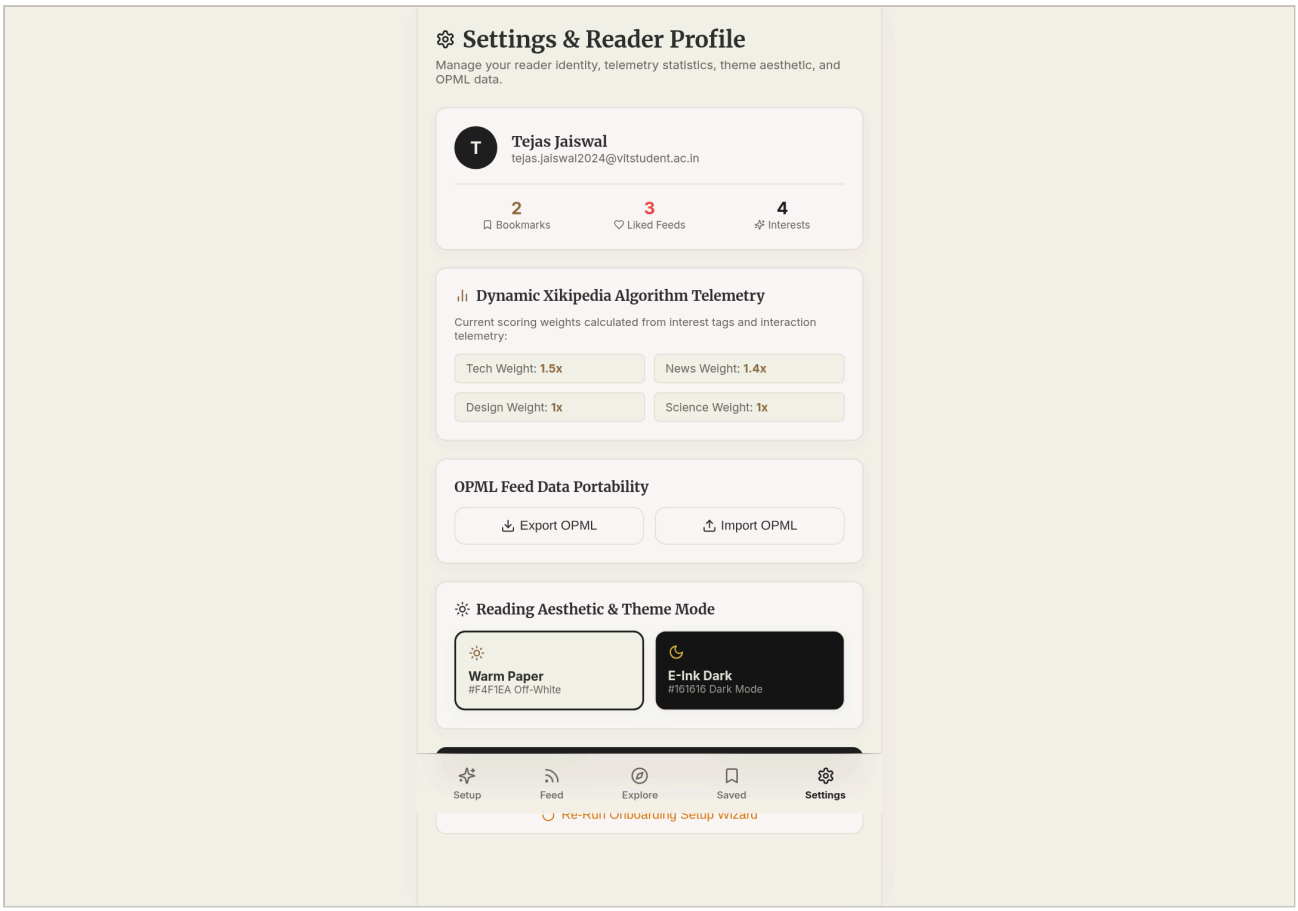


Figure 22 — Settings: reader profile, algorithm weights, OPML export/import, Warm Paper / E-Ink Dark theme (US #5).

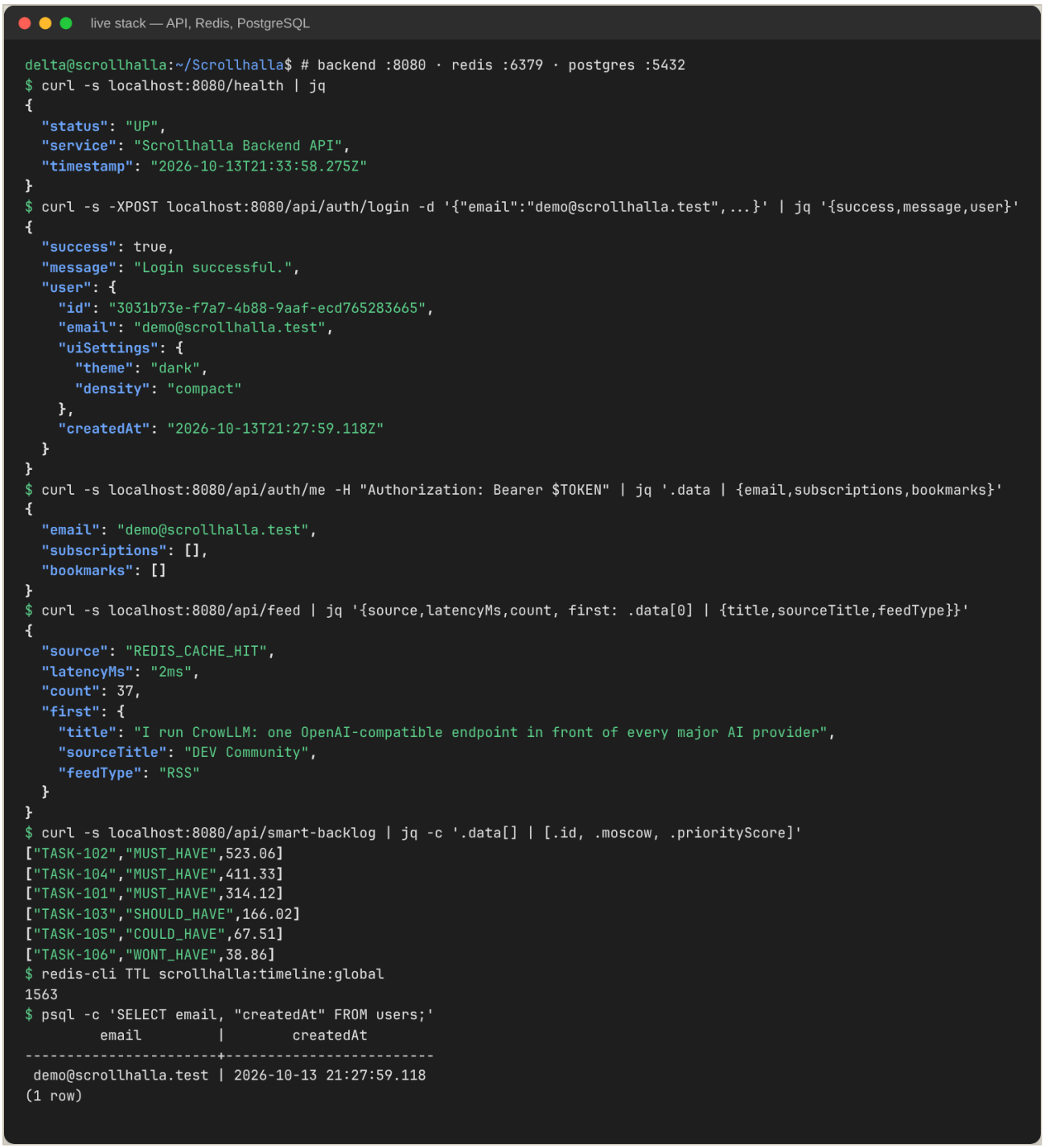


Figure 23 — Backend functionality on the live stack. Health check; login with bcrypt and JWT against PostgreSQL; protected profile; timeline served as `REDIS_CACHE_HIT` in 2 ms; Smart Backlog ranking; Redis TTL; user row in PostgreSQL (US #14, #15, #22).

4. Tools and Technologies Used

Area	Tool / technology	Purpose in Scrollhalla
Frontend	React 18, TypeScript, Vite 5, React Router 6, lucide-react	Single-page e-reader UI, client-side routing, icon set.
Backend	Node.js 20, Express 4, TypeScript 5	REST API, controllers and domain services.
Feed parsing	rss-parser (behind `BaseParser` / `FeedParserFactory`)	RSS 2.0 and Atom 1.0 normalisation to one `IFeedItem` contract.
Scheduling	node-cron	Feed harvest every 15 minutes into Redis.
Database / ORM	PostgreSQL 15, Prisma 5	Users, subscriptions, articles, bookmarks; type-safe queries; schema push.
Cache	Redis 7, ioredis	Pre-computed timeline (TTL 1800 s), with an in-memory fallback.
Security	bcryptjs, jsonwebtoken	Password hashing, stateless JWT authentication.
Unit / integration / regression testing	Jest 29, ts-jest, supertest, Jest snapshots	Backend test pyramid.
Mutation testing	Stryker Mutator 9 (jest-runner)	Test-suite effectiveness, mutation score gate.
Frontend testing	Vitest	OPML service unit tests.
Version control	Git, GitHub (GitHub Flow, PRs, Issues, Projects Kanban)	Branching, code review, user-story tracking.
CI	GitHub Actions	Build, test, mutation gate and container build on every push.
Containers	Docker, Docker Compose, Nginx	Four-service deployment and an `/api` reverse proxy.
Design	Google Stitch, Mermaid	UI screens (DA1); architecture and UML diagrams as code.

5. Demo Plan

The live demo follows the reader's path through the system and then shows the engineering evidence behind it. It takes about 8 minutes.

#	Step	What it demonstrates
1	`docker compose up --build`	Builds the whole system with one command and starts all four containers.
2	Open `localhost:3000` and complete onboarding	Interest tags and seed channels (US #3).
3	Scroll the timeline; like, save and share an article	Reels-style feed served from Redis; overlay actions.
4	Open Focus Reader; change the font size	Distraction-free reading and the accessibility setting.
5	Explore → subscribe by custom URL; Saved tab	Feed discovery and bookmarks.
6	Settings → Export OPML; switch to E-Ink Dark	Feed portability and theming.
7	`curl /api/feed` twice	`ON_DEMAND_HARVEST` on the first call, `REDIS_CACHE_HIT` on the second (inverted read path).
8	`npm test` and `npm run test:mutation`	111 passing tests and a 96.5% mutation score.
9	GitHub → Actions tab, commit graph, tags	The CI run, branching strategy and releases.

6. Problem Statement

Readers face extreme algorithmic noise, cluttered interfaces and information overload in mainstream social media and news apps. Feeds are ranked by engagement rather than by the reader's intent, and long-form text is presented in high-contrast, ad-dense layouts that tire the eye. Standard RSS readers solve the control problem but keep a dated, list-heavy interface that discourages continuous reading.

Scrollhalla is a minimalist, mobile-first RSS/Atom aggregator that combines two things:

- the *control* of RSS, so readers decide which sources they follow;
- the *frictionless discovery* of a vertical, Reels-style scroll.

It presents articles in a calm e-reader aesthetic, with a warm paper background and serif reading text. Feeds are harvested in the background and cached, so opening the app never waits on third-party servers.

Target users.

- *The Deep Thinker* reads long-form essays, tech blogs and news, and wants a paper-like experience without eye strain.
- *The Curator* is tired of algorithms choosing their feed and wants to subscribe explicitly through tags and lists.

Success criteria.

- Feed loads under 500 ms. Measured at 0–2 ms from the Redis cache.
- Standard RSS 2.0 and Atom 1.0 feeds parse correctly. Verified by the parser unit tests.
- The system deploys with a single command through Docker Compose.

7. User Stories

User stories are tracked as GitHub Issues and on the GitHub Projects Kanban board, as set up in DA1. The table maps each story to its MoSCoW priority, its implementation status in this release, and the evidence for it: a test ID from Section 9 or a screenshot.

#	User story	MoSCoW	Status	Evidence
3	As a new user, I want an onboarding screen with interest tags so I can bootstrap my initial feed.	Must	Done	Fig. 16
4	As a user, I want my feed in a vertical, Reels-style scroll so I can move seamlessly between article snippets.	Must	Done	Fig. 17–18
5	As a reader, I want an off-white, paper-like background to reduce eye strain.	Should	Done	Fig. 18, 22
6	As a reader, I want a highly readable serif font for article content.	Should	Done	Fig. 19
7	As a user, I want a “Like” button so the app learns my preferences.	Should	Done	Fig. 18; TC-U-SB
8	As a user, I want a “Save” button on a feed item to read it later.	Should	Done	Fig. 18, 21
9	As a user, I want a Saved/Bookmarks screen listing my saved articles.	Should	Done	Fig. 21
10	As a user, I want an Explore page to discover feeds by topic.	Should	Done	Fig. 20
11	As a user, I want to paste a custom RSS URL on Explore to add a blog.	Should	Done	Fig. 20
12	As a reader, I want to open a snippet in a full-screen “Focus View”.	Should	Done	Fig. 19
13	As a backend system, I want to fetch feeds automatically so content stays fresh.	Must	Done	node-cron every 15 min; TC-I-FEED
14	As a backend system, I want to parse both RSS 2.0 and Atom feeds.	Must	Done	TC-U-PAR, TC-U-FAC
15	As an application, I want to cache parsed feed JSON in Redis for instant load.	Could	Done	TC-I-FEED; Fig. 23
16	As a user, I want to remove a feed from my subscriptions.	Should	Partial	Data model supports it (cascade); no UI yet
17	As a user, I want to adjust the base font size for accessibility.	Could	Done	Fig. 19 (A-/A+)
18	As a user, I want an estimated “Read Time” on each card.	Should	Done	Fig. 17–18
19	As a user, I want to log out securely.	Must	Partial	JWT clear in `apiService`; no UI button
20	As a user, I want a thumbnail next to the article title.	Could	Open	Deferred
21	As a user, I want a minimalist bottom navigation bar (mobile).	Should	Done	Fig. 17
22	As an Admin, I want to configure PostgreSQL via environment variables.	Must	Done	`DATABASE_URL`; Fig. 23
23	As a developer, I want to deploy frontend and backend together with Docker Compose.	Must	Done	CI docker job; §2.2

#	User story	MoSCoW	Status	Evidence
24	As a user, I want an empty state if I have no saved articles.	Could	Done	BookmarksView empty state
25	As a user, I want to share an article link via a “Share” button.	Could	Done	Fig. 18

Summary: **20 of 23** stories are complete, 2 are partial and 1 is deferred. Every *Must Have* that the reader sees is complete. The two partial stories are already supported in the backend and data model and need only a UI control.

8. System Architecture and System Design

8.1 Architectural style

Scrollhalla is a **layered (n-tier) architecture**. It runs as four containers orchestrated by Docker Compose: an Nginx-served React SPA, an Express API, PostgreSQL and Redis. A layered style was chosen over microservices because the deciding constraint is team size, not scale. Two developers would pay the full coordination cost of independent services and gain nothing, since there is no independent scaling requirement.

One refinement is deliberate: the **read path is inverted**. A cron worker harvests the external feeds every 15 minutes and writes a normalised timeline to Redis, so `GET /api/feed` becomes a single key lookup. Freshness is traded for latency and for isolation from third-party servers.

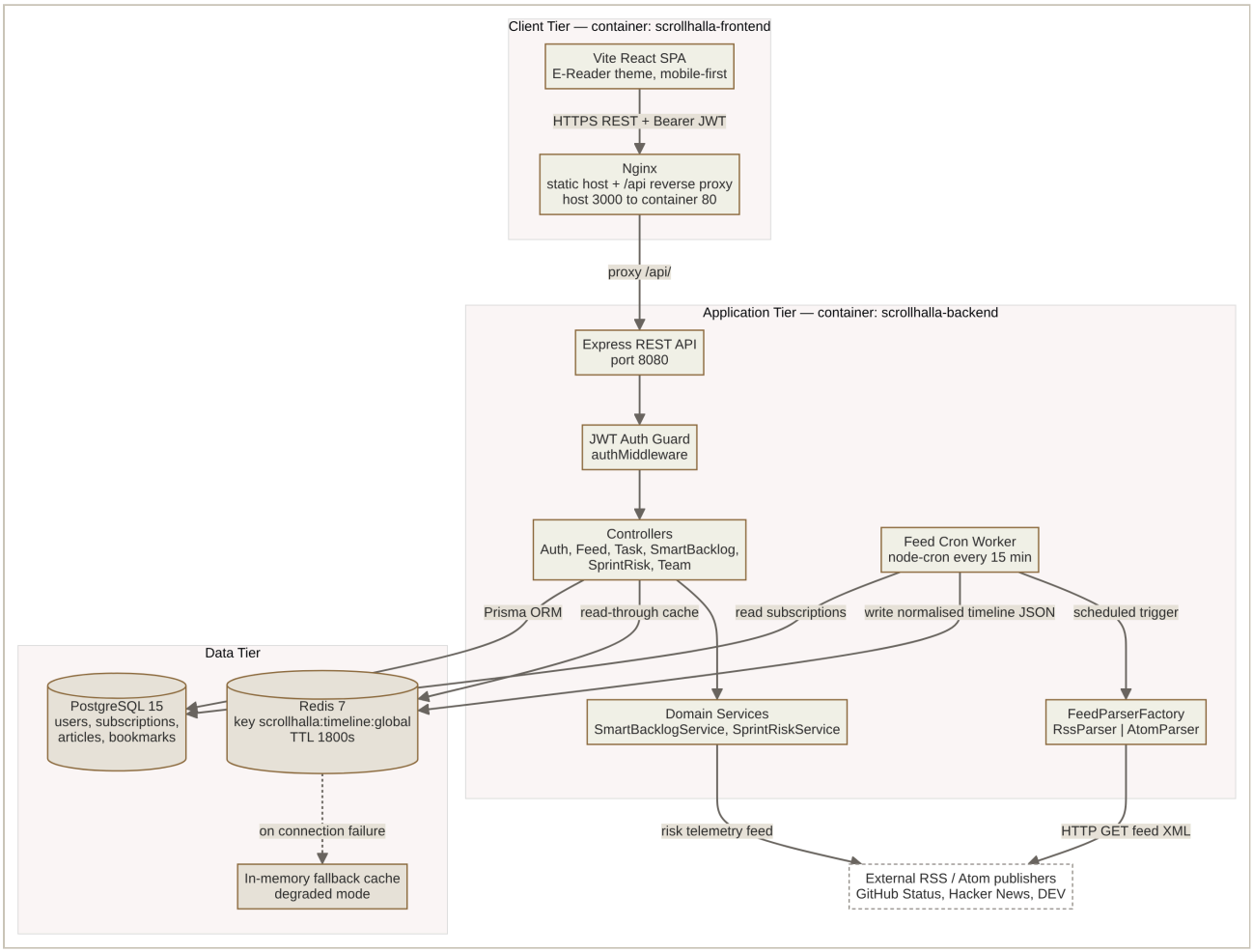


Figure 24 — Architecture and deployment view.

8.2 Design principles and patterns

- **Abstraction and polymorphism.** The abstract `BaseParser` defines one contract, `parse(url): IFeedItem[]`. `RssParser` and `AtomParser` implement it.
- **Factory Method.** `FeedParserFactory` keeps object creation out of the callers. DA3's mutation testing simplified it to a single Atom check (§1.5).
- **Rich domain model.** `Task.getDaysInBacklog()` and `BacklogItem.getMoscowWeight()` keep behaviour next to the data it uses.
- **Strategy and configuration.** `SmartBacklogService` takes injectable weights, which the tests exercise directly.

- **Dependency injection.** `routes/api.ts` is the composition root, and a single `TaskController` is shared by the Backlog and Team controllers.
- **Resilience.** `CacheService` falls back to an in-memory store, and `SprintRiskService` falls back to known telemetry.

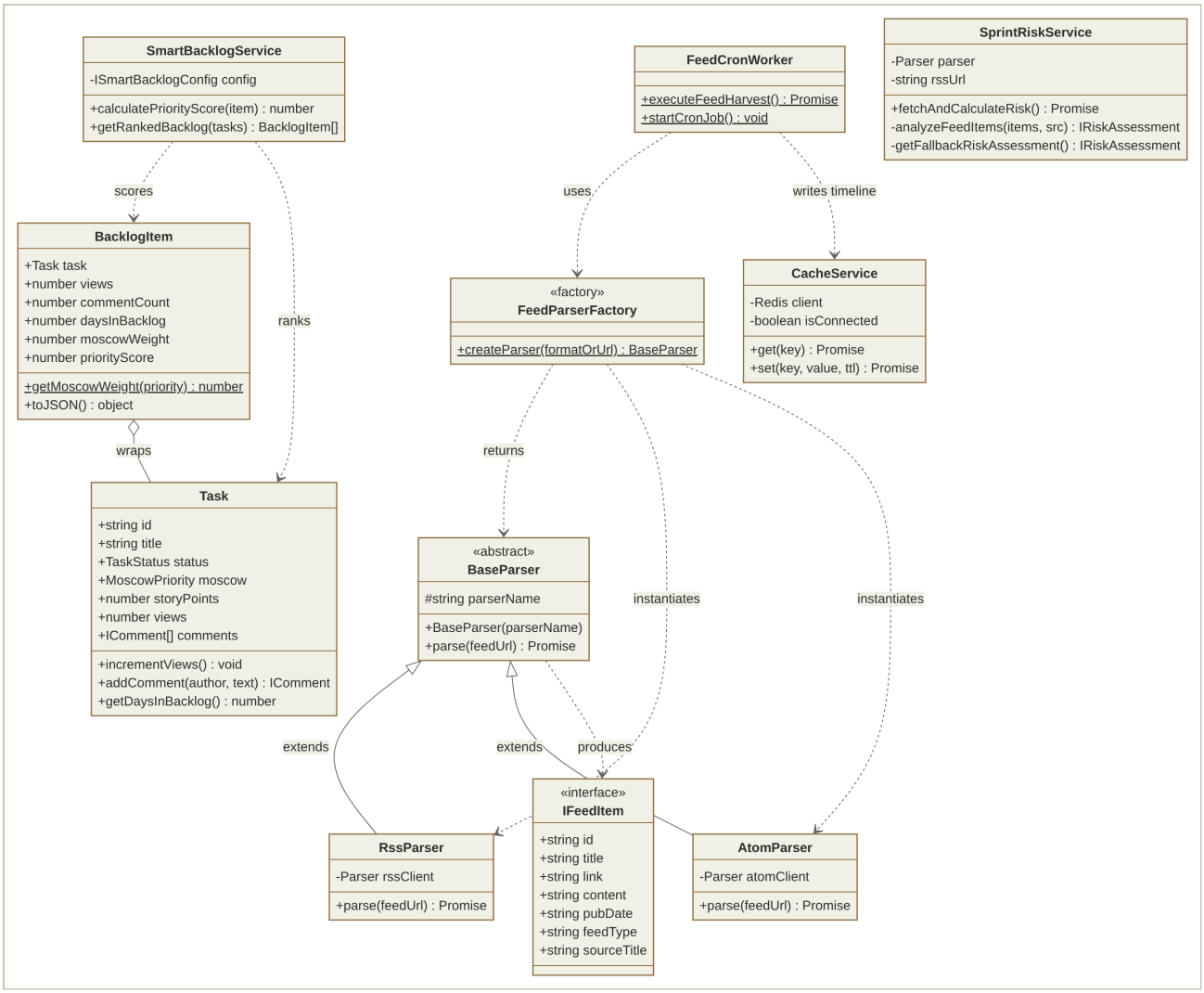


Figure 25 — Class diagram of the backend domain.

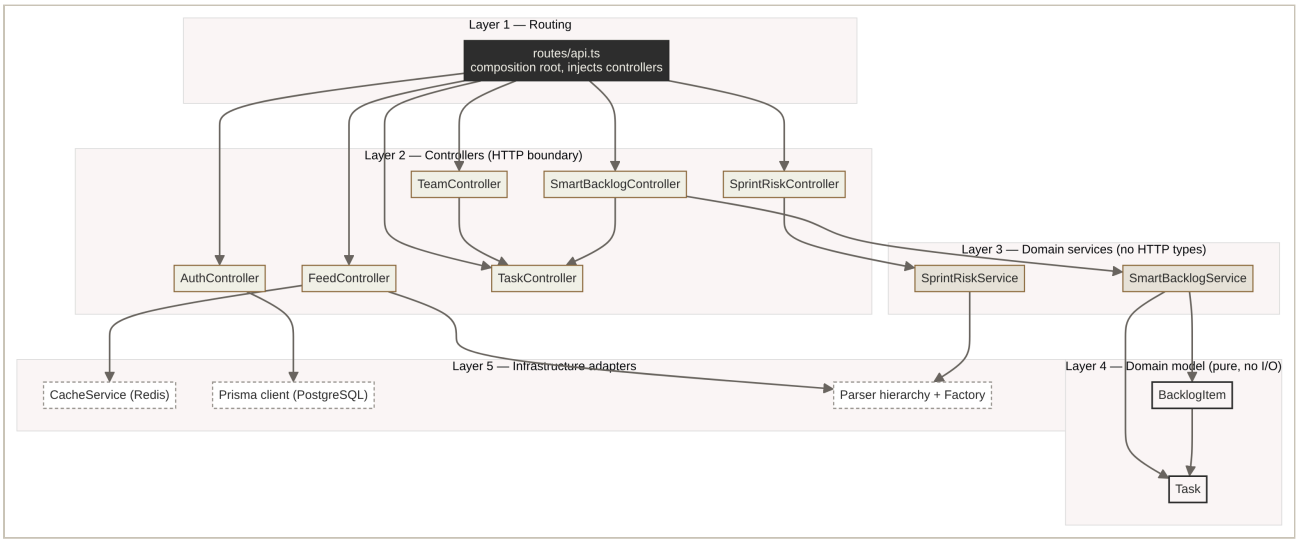


Figure 26 — Five-layer module dependencies. Arrows only point downward.

8.3 Key design decisions

Decision	Why	Cost
Abstract parser behind a factory	A new feed format means one new subclass plus one line in the factory. Callers stay closed to modification.	One extra layer of indirection.
Inverted read path (cron → Redis)	Takes seconds of third-party latency off the request path.	Content can be up to 15 minutes stale.
Cache with in-memory fallback	If Redis is down, performance degrades but the product still works.	Fallback state is per process.
Rich domain classes	The scoring rule cannot drift away from the fields it reads.	Objects must be rehydrated from JSON.
Nginx reverse proxy for `/api`	One origin, so no CORS configuration in production.	Local development needs a matching Vite proxy.

The full design rationale, the six UI screens and the design-system board are in the DA2 Software Design Document (`design/Scrollhalla_DA2_Software_Design_Document.docx`).

9. Design of Tests

9.1 Strategy

The test design follows the architecture's layers. Each level replaces exactly one boundary, which is why every failure points to a single place:

- **Unit tests** isolate a class from all I/O.
- **Integration tests** keep everything real except the Layer-5 adapters: Prisma, Redis and outbound HTTP are swapped for in-memory fakes with the same interface.
- **Regression tests** freeze the externally visible contract.
- **Mutation testing** then measures whether the three levels above would catch a real defect.

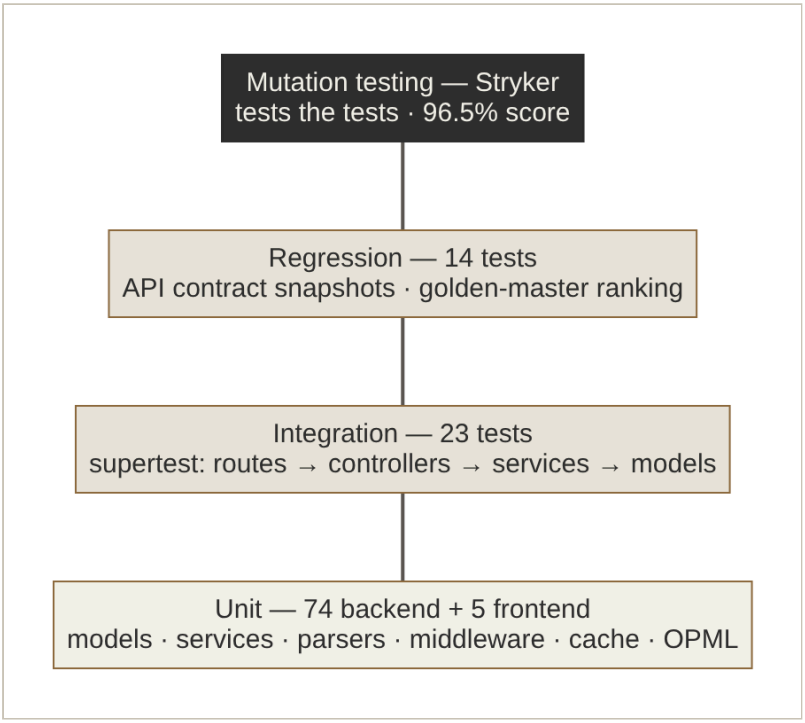


Figure 27 — Test pyramid, with mutation testing on top measuring the levels below.

9.2 Test-design techniques

- **Equivalence partitioning.** MoSCoW priorities (4 valid plus 1 unknown); feed URLs (Atom, RSS, generic); auth headers (missing, wrong scheme, forged, expired, valid).
- **Boundary-value analysis.** Risk thresholds at 19/20, 39/40 and 69/70 and the cap at 100; the risk floor of 5%; the 7-day incident window, exactly at the limit and one second over; `getDaysInBacklog` never below 1; team utilisation capped at 100%.
- **Decision tables.** Severity classification from title and body keywords (Table 9.3, TC-U-SR).
- **State-based testing.** The task lifecycle: create → read (views +1) → update → comment → delete → 404 on every verb.
- **Error guessing and fault injection.** Network failure for each parser and the risk service; one failing publisher among several; Redis unavailable; malformed JSON body.
- **Golden master and contract snapshots.** Used for the regression pack.

9.3 Test cases (representative)

ID	Unit under test	Input / condition	Expected result	Result
TC-U-T1	Task constructor	Only title and description	TO_DO, SHOULD_HAVE, 3 points, “Unassigned”, id `TASK-dddd`	Pass
TC-U-T2	Task.getDaysInBacklog	Created now; created 4.2 days ago	1; 5 (rounded up, never below 1)	Pass
TC-U-B1	BacklogItem.getMoscowWeight	MUST, SHOULD, COULD, WONT, unknown	2.5, 1.8, 1.2, 0.5, 1.0	Pass
TC-U-SB1	SmartBacklogService	views 10, comments 2, 4 days, 5 points, MUST	(15 + 8 + 25·ln 5 + 6) × 2.5	Pass
TC-U-SB2	SmartBacklogService	Negative view weight	Score clamped to 0	Pass
TC-U-SB3	getRankedBacklog	Low, high and mid tasks in that order	HIGH, MID, LOW, non-increasing	Pass
TC-U-SR1	SprintRiskService severity	“major” / “outage” / body “major outage”	CRITICAL (30)	Pass
TC-U-SR2	SprintRiskService severity	“degraded” / “partial” / body “incident”	HIGH (20)	Pass
TC-U-SR3	SprintRiskService thresholds	Weights summing to 15, 20, 40, 70, 120	LOW, MEDIUM, HIGH, CRITICAL, capped at 100	Pass
TC-U-SR4	SprintRiskService window	Incident exactly 7 days old, and 7 days + 1 s	Counted, not counted	Pass
TC-U-SR5	SprintRiskService fallback	Feed unreachable	18%, LOW, two past incidents, “(Resilience Fallback Mode)”	Pass
TC-U-PAR1	RssParser	Full RSS item	Normalised `IFeedItem`, ISO date, `feedType` RSS	Pass
TC-U-PAR2	AtomParser	Entry with summary and updated	Content from summary, date from updated, ATOM	Pass
TC-U-PAR3	Rss/AtomParser	Network error	Resolves to `[]` without throwing	Pass
TC-U-FAC	FeedParserFactory	ATOM, `.atom`, `/atom`, RSS, `.rss`, `/feed`, AUTO	AtomParser × 4, RssParser × 5	Pass
TC-U-MW	authenticateJWT	No header; Basic; forged; expired; valid	401; 401; 403; 403; `next()` with `req.user`	Pass
TC-U-C1	CacheService	Redis down; set, get, TTL expiry	Value round-trips; `null` after TTL	Pass
TC-I-TASK	/api/tasks (CRUD)	Create → get ×2 → update → comment → delete	201; views +1; partial update; 201; then 404 on every verb	Pass
TC-I-DI	/api/smart-backlog + /api/team	New task with 80 views and 20 comments; reassignment	Ranked first; member workload updated	Pass

ID	Unit under test	Input / condition	Expected result	Result
TC-I-FEED	/api/feed pipeline	Cold cache, then a second call; one source failing	ON_DEMAND_HARVEST, newest first; then CACHE_HIT with no refetch; 2 of 3 sources still served	Pass
TC-I-AUTH	/api/auth/*	Register, duplicate, bad login, login, /me with and without token	201 (no hash in body); 409; 401 with identical messages; 200; 200 / 401	Pass
TC-R-01	All endpoints	GET on 9 routes	Status and response shape match snapshot	Pass
TC-R-02	Seed backlog	GET /api/smart-backlog	Order 102, 104, 101, 103, 105, 106	Pass
TC-R-03	Edge behaviours	storyPoints 0; utilisation over 100%; malformed JSON	3; capped at 100; handled error with success: false	Pass
TC-F-OPML	generateOpmlXml	Title `Tom & "Jerry's" <feed>`	All five XML entities escaped; valid OPML 2.0	Pass

9.4 Test doubles

Double	Replaces	Used by
`rssParserMock`	The rss-parser network client, with a programmable `parseURL`	Parser, risk service, integration and regression tests
`prismaModuleMock`	The Prisma client, backed by an in-memory `users` array	Integration and regression tests (auth)
`cacheModuleMock`	The Redis `CacheService`, backed by a `Map`	Integration and regression tests (feed)
Fake timers	The system clock	TTL expiry and 7-day boundary tests

9.5 Exit criteria

- All unit, integration and regression tests pass in CI.
- Mutation score is at least 60% (break threshold). The target band is 70–85% or above; achieved: 96.49%.
- Every surviving mutant is either killed or documented as equivalent (§1.5).
- The backend type-checks (`tsc`) and the frontend builds. The compose stack builds.

Appendix A — Software Requirements Specification

A.1 Purpose and scope

This SRS specifies Scrollhalla, a web-based RSS/Atom e-reader with a Reels-style reading feed. It is delivered as a React SPA and a REST API, deployed with Docker Compose. Out of scope for this release are full offline support, native iOS/Android apps and AI-generated summaries (the *Won't Have* items in DA1).

A.2 Overall description

- **Users:** the Reader (Deep Thinker and Curator personas) and the Admin/DevOps role. A cron scheduler and external publishers are system actors.
- **Operating environment:** a modern browser (desktop and mobile), Node.js 20, PostgreSQL 15, Redis 7, and Docker.
- **Constraints:** deployable with Docker Compose; configuration only through environment variables; no secrets committed.
- **Assumptions:** users have standard RSS/Atom URLs or use the in-app directory; publishers serve valid XML over HTTP(S).

A.3 Functional requirements

ID	Requirement	Priority	Story
FR-1	The system shall let a user register and log in with email and password, store a bcrypt hash, and issue a JWT valid for 7 days.	Must	#19
FR-2	The system shall reject protected requests without a valid Bearer token (401 when missing, 403 when invalid or expired).	Must	#19
FR-3	The system shall provide onboarding where the reader enters a profile, selects interest tags and chooses seed channels.	Must	#3
FR-4	The system shall parse RSS 2.0 and Atom 1.0 feeds into one normalised article format.	Must	#14
FR-5	The system shall harvest all subscriptions every 15 minutes and cache the merged, newest-first timeline for 30 minutes.	Must	#13, #15
FR-6	On a cache miss, the system shall harvest on demand. A failing publisher shall not prevent the other publishers' articles from being served.	Must	#15
FR-7	The system shall present the timeline as a vertical, one-card-per-article scroll with read time, and Like, Save, Read, Share and Open actions.	Must	#4, #7, #8, #18, #25
FR-8	The system shall provide a full-screen reader view with adjustable font size.	Should	#12, #17
FR-9	The system shall list bookmarked articles with search and removal, and show an empty state when there are none.	Should	#9, #24
FR-10	The system shall offer a categorised feed directory and subscription by custom URL.	Should	#10, #11
FR-11	The system shall export subscriptions as OPML 2.0 with XML-escaped fields.	Could	—
FR-12	The system shall offer Warm Paper and E-Ink Dark themes.	Could	#5
FR-13	The system shall let a user remove a subscription.	Should	#16
FR-14	The system shall rank backlog items by views, comments, age, story points and MoSCoW weight (Smart Backlog).	Should	—

ID	Requirement	Priority	Story
FR-15	The system shall compute a 0–100% sprint risk score from the GitHub status feed, with an offline fallback.	Could	—

A.4 Non-functional requirements

ID	Category	Requirement	Verification
NFR-1	Performance	Timeline API responds in under 500 ms when the cache is warm.	Measured 0–2 ms (Fig. 23)
NFR-2	Availability	If Redis is unavailable, the system degrades to an in-memory cache instead of failing.	TC-U-C1
NFR-3	Security	Passwords are never stored or returned in plain text. Login errors do not reveal whether an account exists.	TC-I-AUTH, TC-R-03
NFR-4	Usability	Warm paper palette (#F4F1EA / #2D2D2D); one primary action per screen; navigation within thumb reach on mobile.	Fig. 16–22
NFR-5	Portability	The whole stack starts with `docker compose up`. All configuration comes from the environment.	CI docker job
NFR-6	Maintainability	Adding a feed format needs no change to callers. Mutation score stays at or above 60% in CI.	TC-U-FAC; Stryker gate
NFR-7	Interoperability	Subscriptions can be exchanged with other readers through OPML 2.0.	TC-F-OPML

A.5 External interfaces

Method	Endpoint	Description	Auth
GET	`/health`	Service liveness	—
POST	`/api/auth/register`, `/api/auth/login`	Create account / obtain JWT	—
GET	`/api/auth/me`	Profile with subscriptions and bookmarks	JWT
GET	`/api/feed`	Cached combined timeline (source, latency, count, data)	—
GET/POST/PUT/DELETE	`/api/tasks[/:id[/comments]]`	Backlog task CRUD and comments	—
GET	`/api/smart-backlog`, `/api/sprint-risk`, `/api/team`	Ranked backlog, risk telemetry, team workload	—

Appendix B — Data Flow Diagrams

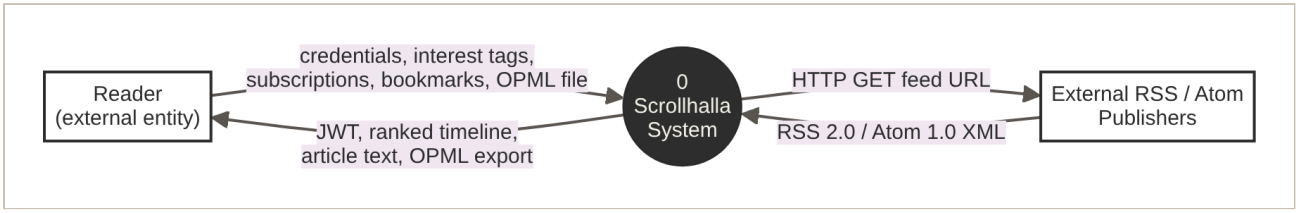


Figure B1 — DFD Level 0 (context diagram).

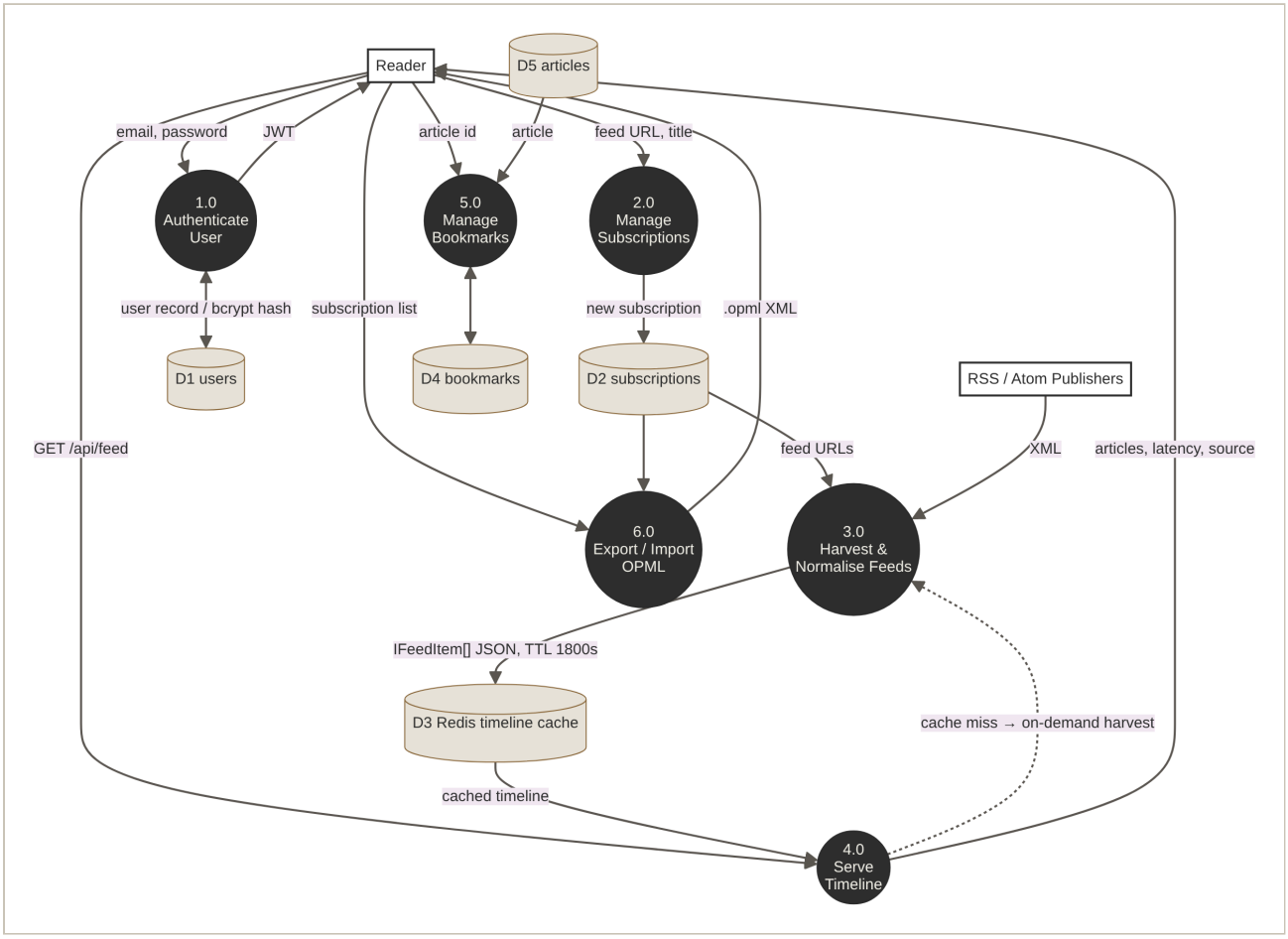


Figure B2 — DFD Level 1: six processes and five data stores. The dotted flow is the cache-miss path.

Appendix C — Entity Relationship Diagram

The ERD is derived directly from `backend/prisma/schema.prisma`:

- a User has many Subscriptions and many Bookmarks;
- a Subscription has many Articles;
- a Bookmark links one User to one Article, and the pair `(userId, articleId)` is unique.

All foreign keys cascade on delete.

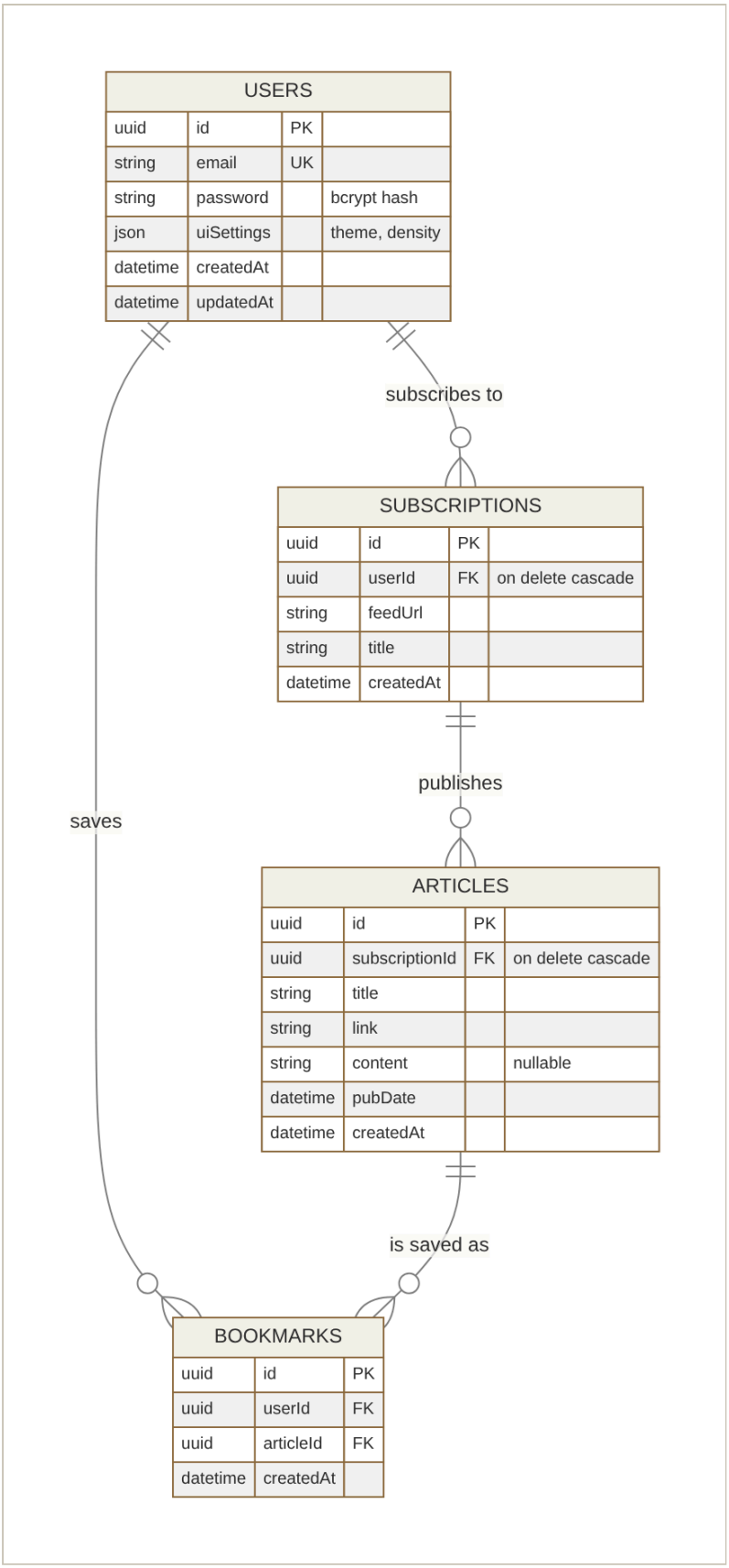


Figure C1 — Entity relationship diagram (PostgreSQL via Prisma).

Appendix D — UML Diagrams

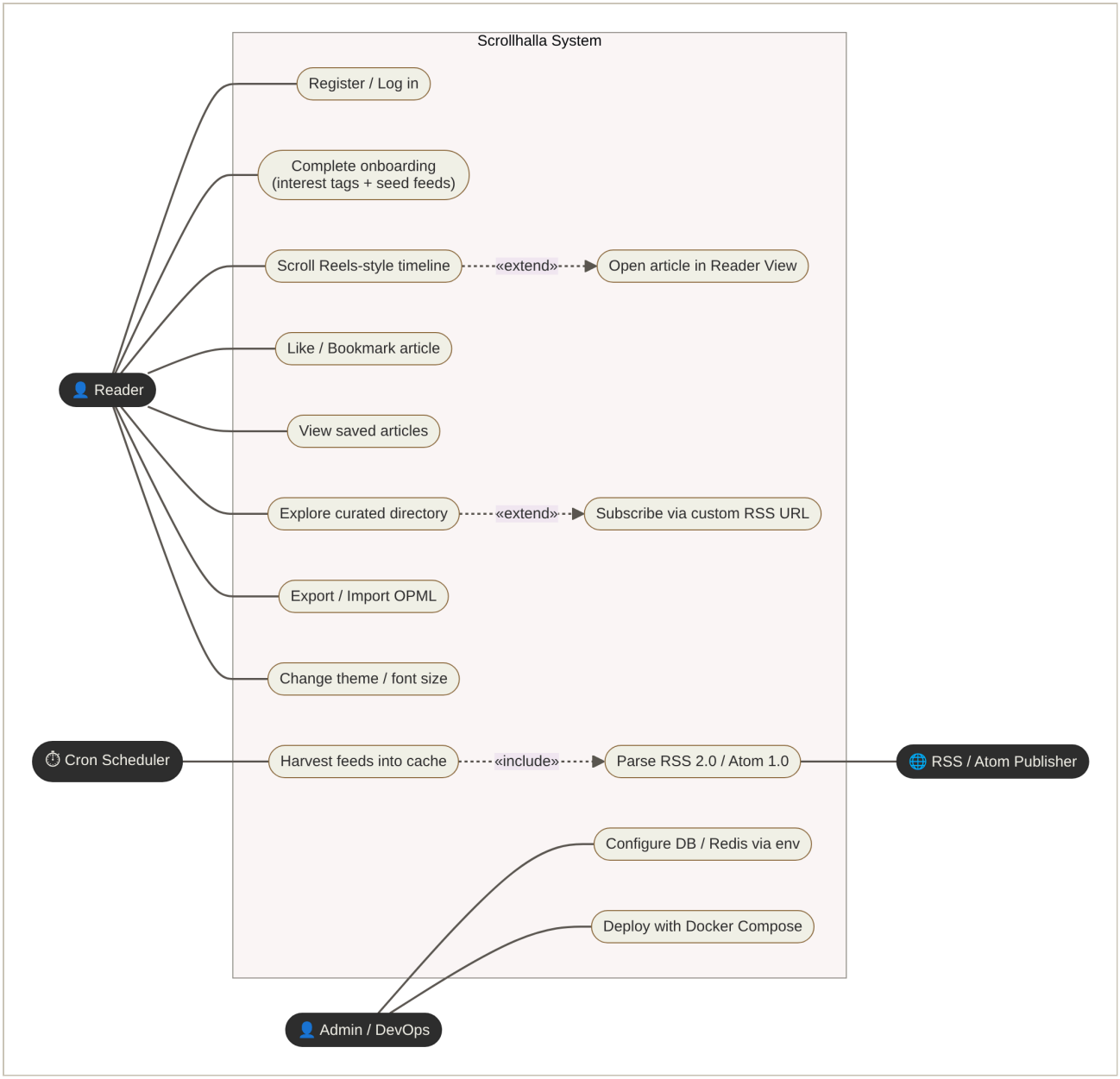


Figure D1 — Use case diagram.

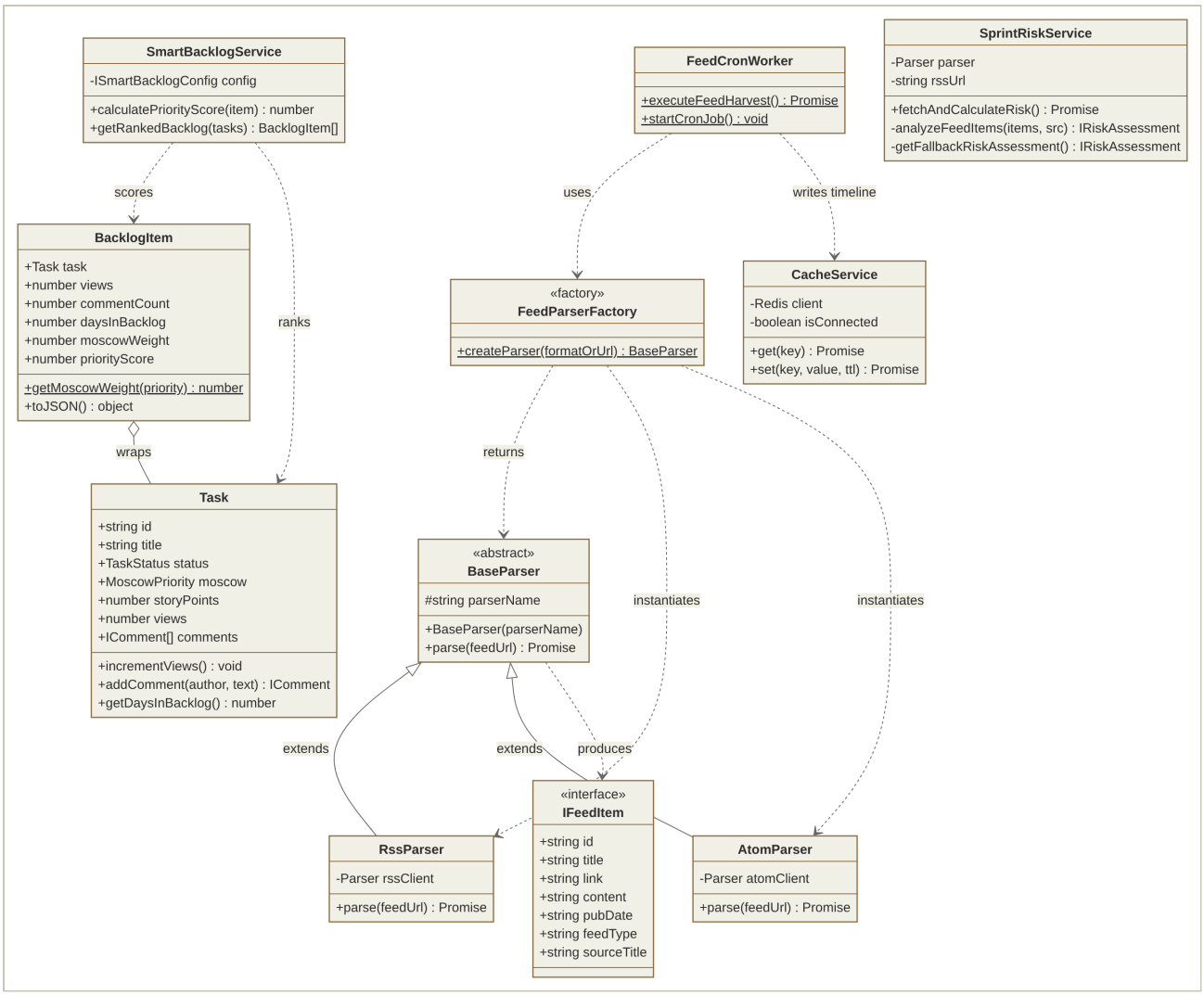


Figure D2 — Class diagram.

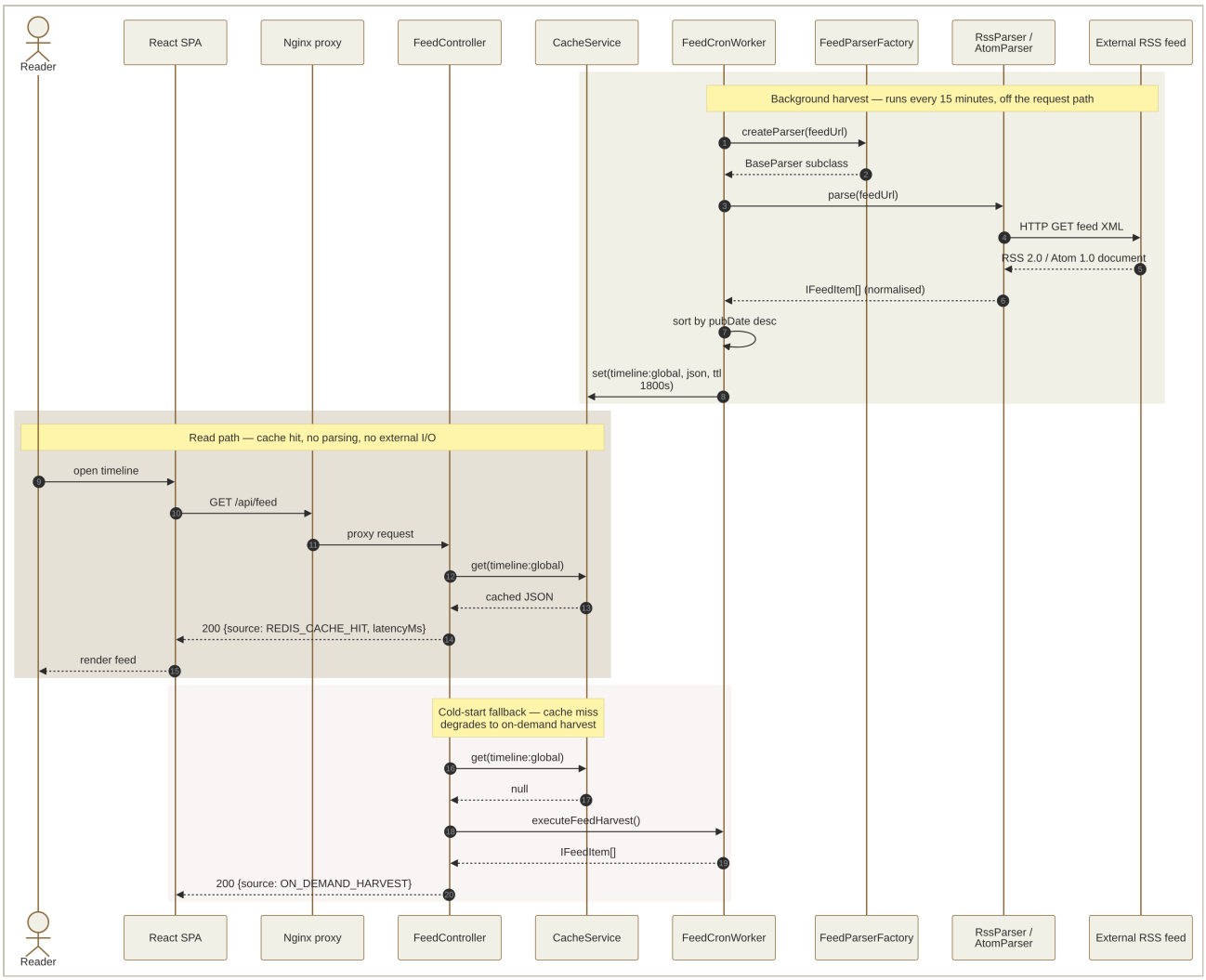


Figure D3 — Sequence diagram: cron harvest, cached read and cold-start fallback.

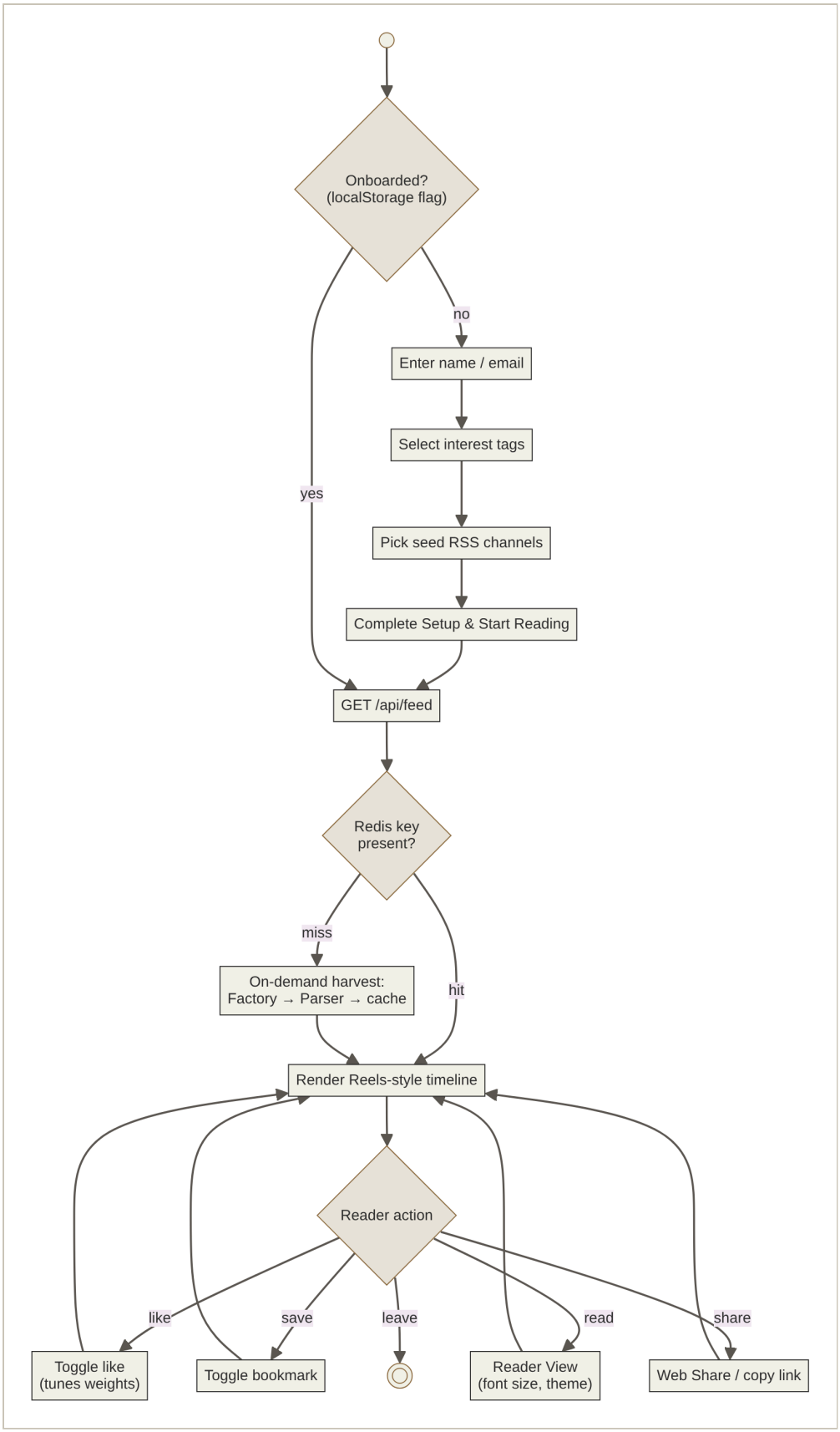


Figure D4 — Activity diagram: onboarding and reading session.

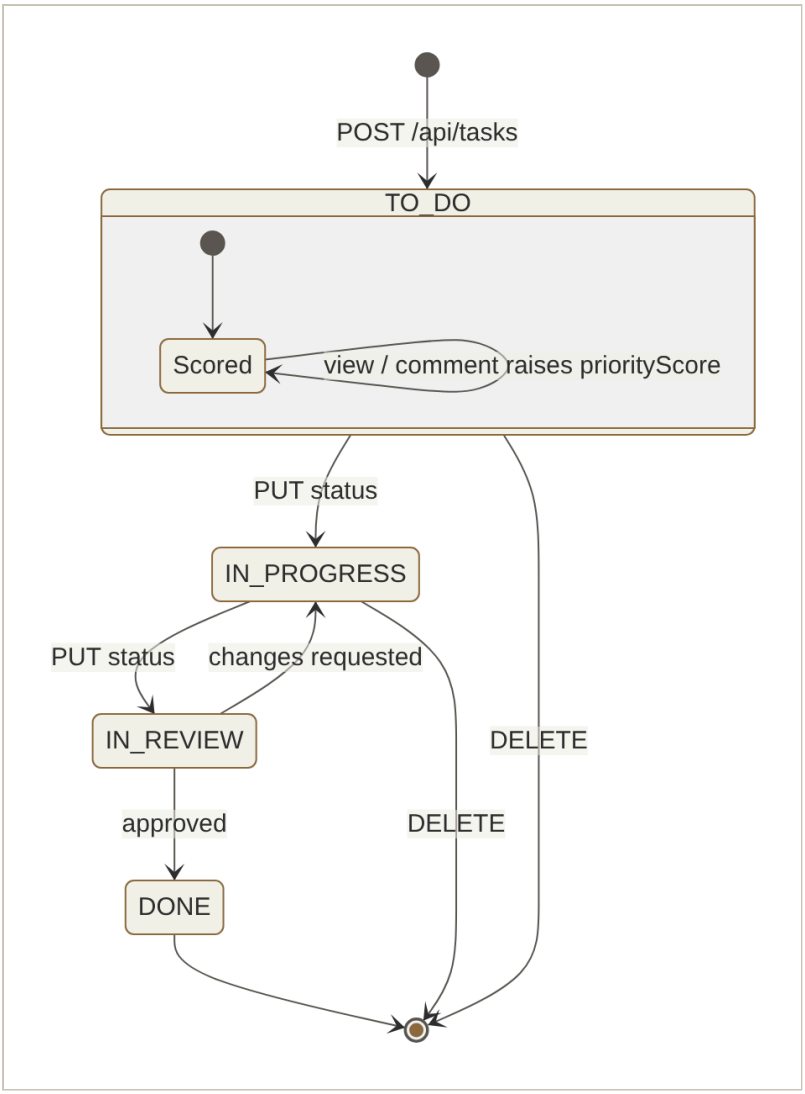


Figure D5 — State machine diagram: backlog task lifecycle.

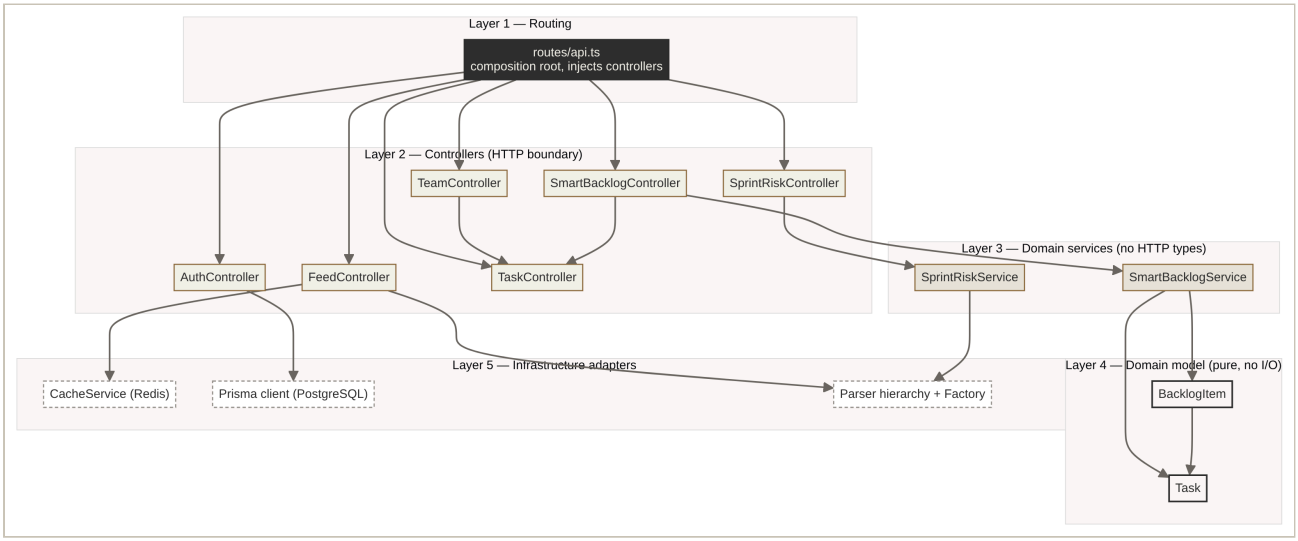


Figure D6 — Component/package diagram: layered module dependencies.

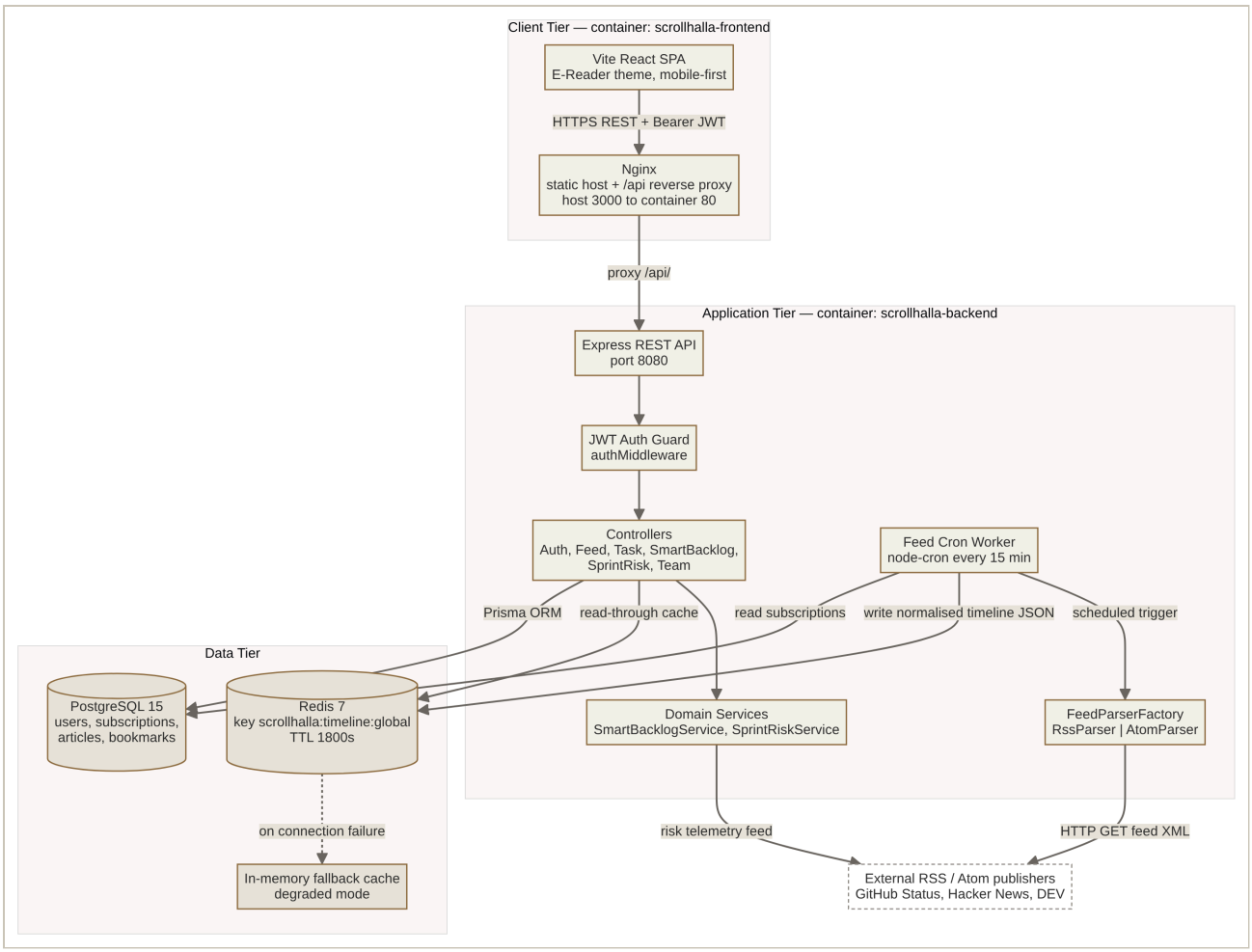


Figure D7 — Deployment diagram: four Docker Compose containers.

Appendix E — Code Listing and GitHub Link

Repository: https://github.com/tejasjaiswal2024-art/Scrollhalla
DA3 branch: `feature/da3-testing` · **Release tag:** `v3.0.0`

Path	Contents
`backend/src/`	Express app, controllers, services, models, parsers, middleware, cron worker, Prisma/Redis config
`backend/prisma/schema.prisma`	Database schema (Appendix C)
`backend/tests/unit|integration|regression/`	Jest suites; `tests/helpers/` holds the test doubles
`backend/jest.config.js`, `stryker.config.json`	Test runner and mutation configuration
`frontend/src/`	React views, components, services (`apiService`, `opmlService`)
`.github/workflows/ci.yml`	CI pipeline
`docker-compose.yml`, `*/Dockerfile`, `frontend/nginx.conf`	System build and deployment
`design/`	Mermaid sources, PNG exports, UI screens, DA2 SDD
`docs-da3/`	This report and all screenshots

Listing E1 — Factory Method after the mutation-driven refactor (backend/src/parsers/FeedParserFactory.ts)

```
export class FeedParserFactory {
  public static createParser(formatOrUrl: string = 'AUTO'): BaseParser {
    return FeedParserFactory.isAtom(formatOrUrl) ? new AtomParser() : new RssParser();
  }
  private static isAtom(formatOrUrl: string): boolean {
    return formatOrUrl.toUpperCase() === 'ATOM' || formatOrUrl.includes('.atom') ||
formatOrUrl.includes('/atom');
  }
}
```

Listing E2 — Smart Backlog scoring (backend/src/services/SmartBacklogService.ts)

```
public calculatePriorityScore(item: BacklogItem): number {
  const viewScore = item.views * this.config.weightViews;
  const commentScore = item.commentCount * this.config.weightComments;
  // Logarithmic decay/growth for time in backlog to avoid extreme skew
  const ageScore = Math.log(item.daysInBacklog + 1) * 10 * this.config.weightBacklogAge;
  const complexityScore = item.storyPoints * this.config.weightStoryPoints;
  const finalScore = (viewScore + commentScore + ageScore + complexityScore) *
item.moscowWeight;
  item.priorityScore = Math.max(0, finalScore);
  return item.priorityScore;
}
```

Listing E3 — Read path served from Redis (backend/src/controllers/FeedController.ts, excerpt)

```
let cachedTimeline = await cacheService.get(REDIS_FEED_CACHE_KEY);
if (cachedTimeline) {
  articles = JSON.parse(cachedTimeline);
} else {
  // Cache miss: execute on-demand harvest & populate cache
  articles = await FeedCronWorker.executeFeedHarvest();
}
res.json({ success: true, source: cachedTimeline ? 'REDIS_CACHE_HIT' : 'ON_DEMAND_HARVEST',
latencyMs, count: articles.length, data: articles });
```

Listing E4 — Mutation testing configuration (backend/stryker.config.json)

```
{
  "testRunner": "jest",
  "jest": { "configFile": "jest.config.js", "enableFindRelatedTests": false },
  "coverageAnalysis": "perTest",
  "mutate": ["src/models/**/*.ts", "src/services/**/*.ts", "src/parsers/**/*.ts",
             "src/middleware/**/*.ts", "src/controllers/TaskController.ts",
"src/controllers/TeamController.ts"],
  "thresholds": { "high": 85, "low": 70, "break": 60 }
}
```

Conclusion

Review 3 takes Scrollhalla from an untested codebase to one with a four-level automated test suite of 116 tests, 92% line coverage and a 96.5% mutation score, enforced by a CI pipeline that builds and tests every push. Mutation testing was the most informative of the levels. Besides measuring the suite, it found dead code in the parser factory and three blind spots that coverage figures had hidden. The system builds and runs end-to-end on PostgreSQL and Redis with live feeds. Twenty of twenty-three user stories are complete, and the remaining three are recorded with their current state.